# 14. Research topics

*What the model opens for other researchers*

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**Abstract**

• **The mechanism discussed in the series:** The LEF model (Lunar Elevation and Fission) proposes that the Moon was built from the Earth's own material while it was still attached to it. A hot, fast-spinning Earth raised a pear-shaped extension of its own material; deep, volatile-rich melt was fed up through it into a body on its tip; the Earth's outer shell over the region rose with that body as the surface of a dome and became the Moon's crust; and the body was released when its centre of mass reached the corotation line (the distance at which a body turning synchronously with the Earth moves at free orbital speed). Papers 1–13 of the series set out this process, its orbit, the maria, the craters, the poles, the South Pole–Aitken basin, the stripes of the Pacific floor, the moons of Mars and the requirements the Moon must meet.

• **The role of Paper 14:** This companion paper performs no new calculations. It collects, from all 12 previous papers, every topic that has been left open: requirements not yet tested, studies defined but not yet run, calculations marked "to be computed", and rows that the summary table of Paper 1 marks as partial or awaiting. To these are added 15 questions from the working notes of the series (9 of them the author's own proposals that the papers have not yet taken up). Recurring items have been merged into a single topic, with all sources kept.

• **Structure of the list:** The paper gathers 132 research topics, grouped by theme in 11 parts (A–K). Part L gives three lists for choosing: the two key observations of the model, the topics that can be done at once on public data, and the topics that need a large simulation, a space mission or laboratory samples.

**What this paper proposes (and its limits)**

• **Distributed calibration of the model:** The model is fully stated; its calibration requires broad computational and observational tools (three-dimensional simulations, equations of state at $50 - 150\ \mathrm{GPa}$, coupled thermal–orbital integrations, laboratory work, space missions and samples). Each requirement of Paper 13 is defined as a topic

for a paper of its own, with its own test, the deciding observation and the expected numerical value.

• **Kinds of topics and the key observations:**

  • **Calibration topics:** sharpening parameters that the series gives as a range (the expansion of the hot parent, the time of failure of the contact, and the dissipation history of the Earth).

  • **Topics of partial confirmation:** extending predictions already partly supported (the order of sealing of the maria, the young craters of the poles, and the LLVP province beneath the Pacific).

  • **The two key observations of the model:**

  **1. The geometry of the exposure zones (topics 121–122):** whether the crust and the crater record are ordered in three concentric zones about the centre of the near side.

  **2. The source of the craters (topics 49 and 123):** whether the material that made the craters is returned terrestrial material (as the model requires) or foreign material from outside the Earth–Moon system.

• **What this paper does not do:** The paper recomputes no results, changes no numerical value of the series and weakens no claim; every number quoted is given with a reference to the original paper and section. Two subjects named in the series as future work — the Earth–Moon axes system and the Moon's magnetism — are noted at the end of the paper and are not counted among the 132 topics.

## Introduction — how to use this paper

**1. Definition of a research topic and its five sources**

A research topic is defined as one open question of the model that merits a paper of its own, and it is drawn from one of five sources:

**1.** A requirement that has been stated but not yet tested (the requirements R1–R18 of Paper 13).

**2.** A study defined in a paper with thresholds fixed in advance, but not yet run.

**3.** A calculation marked in the papers as still to be made.

**4.** A row of the summary table of Paper 1 (§D.1), which marks each claim as shown, partial or awaiting (in traffic-light colours: green, orange and red).

**5.** A question from the working notes of the series, with the model's expected answer, that no paper has yet taken up (15 topics at the end of Parts B, D, E, G, I and K).

### 2. How each topic is presented

Each topic opens with a short paragraph that explains what the question is about and why it matters to the model, followed by five fixed tagged lines:

- **The question:** the problem stated in one sentence.
- **What the model predicts:** the numerical value or the direction the series expects, with a reference to the relevant paper.
- **Data:** the data sets that exist or are needed.
- **Method:** the computational, observational or laboratory tool required.
- **Where in the series:** an exact reference to the original paper and section.

**3.** The parts and their topics (A–L)

| Part | Main theme | Number of topics | Numbering |
|---|---|---|---|
| A | The Earth before and during the pear | 11 | 1–11 |
| B | The body on the tip — shape, rounding, release | 12 | 12–23 |
| C | The orbit | 6 | 24–29 |
| D | The maria and the openings | 14 | 30–43 |
| E | The craters and their source | 14 | 44–57 |
| F | The poles | 10 | 58–67 |
| G | South Pole–Aitken | 12 | 68–79 |
| H | The stripes and the Pacific | 11 | 80–90 |
| I | Mars and other bodies | 16 | 91–106 |
| J | Moon births across the solar system | 14 | 107–120 |
| K | Requirements and key tests | 12 | 121–132 |
| L | How to choose — three lists | — | — |
| Total | — | 132 | 1–132 |

### 4. How ready the topics are

| Readiness | Practical meaning for the researcher |
|---|---|
| Public data | The study can be done at once on data already published (catalogues, maps, gravity and topography grids, and existing isotope data). |
| New modelling | The data exist or none are needed, but a new analytical calculation or numerical simulation must be developed. |
| New data | The study needs a dedicated space mission, returned samples, laboratory experiments or new observations. |

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PART A

# The Earth before and during the pear

Part A covers the parent body: the early Earth before the Moon separated from it. In the model the parent was hot, rich in volatiles, completed one rotation in about 5.5 hours, and was in the midst of differentiating and separating its iron toward the centre. From one region of it (near the island of Kiritimati in the central Pacific), a pear-shaped extension of the Earth's own material rose toward the corotation line, and the Moon was built on its tip. Topics 1–11 set the dynamical and thermodynamic starting conditions of the system (ten of them need new modelling and one needs new laboratory data).

### Topics of Part A

| No. | Topic | Main tool | Readiness |
|---|---|---|---|
| 1 | Thermal expansion of the hot parent | Equation of state | New modelling |
| 2 | The stratified Maclaurin problem | Figure of a rotating body | New modelling |
| 3 | Core formation and the Hf–W history | Reservoir model | New modelling |
| 4 | Equation of state of the deep melt | Experiments, ab-initio | New data |
| 5 | Energy of the lift and of the parent | Thermal evolution | New modelling |
| 6 | Pressure test with the pear and flattening | Non-spherical field | New modelling |
| 7 | Construction rate: the feed flux | Pipe flow and freezing | New modelling |

| | | | |
|---|---|---|---|
| 8 | The pear’s equilibrium figure (R9) | Free-surface figure | New modelling |
| 9 | The northward lean of the pear | Feed-path geometry | New modelling |
| 10 | The Earth’s core during the pear | 3-D hydrostatic model | New modelling |
| 11 | The classical fission route, recomputed | Fluid equilibria | New modelling |

## 1 The thermal expansion of the hot parent

A hot body saturated with dissolved volatiles is larger in volume than a cold one; at a given mass, the radius of the young Earth sets the window of rotation periods in which the transfer mechanism can work. Paper 2 adopts an expansion of 120 km and shows that the model works over a range of values; a full calculation would narrow that range.

- **The question:** How much larger in radius was the hot, volatile-rich early Earth than a cold one?
- **What the model predicts:** An expansion of 120 km is adopted (admissible range 98 - 520 km), which sets a window of rotation periods for the transfer of 4.90 - 5.54 hours at $\Delta R$ = 120 km (Paper 2).
- **Data:** Estimates of the depth of the early magma ocean; thermal expansion coefficients of silicate melts.
- **Method:** An integration of an equation of state (which relates pressure, temperature and density) from the centre outward for a hot, volatile-rich Earth.
- **Where in the series:** Paper 2, Further work item 3; §D.3.

## 2 The stratified Maclaurin problem

A Maclaurin spheroid describes a uniform fluid body in steady rotation, whereas the early Earth was undergoing iron descent that made its interior denser than its mantle. The position of the corotation line (the Moon’s starting orbit) depends on the distribution of density inside.

- **The question:** Where does the corotation line lie for a parent whose iron is sinking, computed self-consistently?
- **What the model predicts:** A range of 15,382 - 17,569 km from D = 3 to D = 1 (where D is the ratio of the density of the inner layer to that of the outer layer; D = 1 is a uniform body). At D = 2 the line lies at 16,300 - 16,400 km, which is the starting orbit of Paper 4 (Paper 2).

• **Data:** The parent parameters of the series: volumetric radius 6,517 km, rotation period 5.5 hours, and moment-of-inertia factor k $\gtrsim$ 0.397 (where k = 0.4 for a uniform sphere and less for a body with a dense core).

• **Method:** The Clairaut–Darwin–Radau theory of the figure of a rotating body, applied to a body with active internal differentiation.

• **Where in the series:** Paper 2, Further work item 1; §A.4.1, §A.6.2; Paper 1 §F.2.

## 3 Core formation during the transfer, and the Hf–W reservoir history

The short-lived radioactive isotope $^{182}$Hf (a lithophile that stays in the silicate) decays to $^{182}$W (a siderophile that follows iron into the core), so an excess of $^{182}$W in a reservoir records when its metal separated. The model requires that the Earth's core was still forming during the transfer and that one chronology carry the tungsten signatures through the whole window of construction.

• **The question:** Was the Earth's iron still descending during the transfer (centralisation D ≈ 2), and can one chronology carry the tungsten signatures through a 600-Myr window?

• **What the model predicts:** An essentially uniform parent at the start (k ≥ 0.397); the centralisation of the core during the transfer shortens the day at release from 6.44 to 5.77 hours (Paper 2), and the Hf–W excess was set early in the window and carried without reset (Paper 13).

• **Data:** $^{182}$W anomalies in terrestrial and lunar samples; Hf/W ratios.

• **Method:** A reservoir model of metal–silicate separation over a 600-Myr window under partial equilibration of metal and silicate.

• **Where in the series:** Paper 2, requirements box of Part D, §D.3; Paper 13 §D.2; Paper 1 §F.2 and §D.1 (the orange row of Paper 2).

## 4 The equation of state of the deep, volatile-rich melt at $50 - 150$ GPa

The pressure test of Paper 3 asks whether the deep source could push its melt up through the pear and into the body on its tip; the answer depends on the density of a melt rich in water and carbon dioxide at the pressures of the deep mantle ($50 - 150$ GPa), since a lighter melt needs less pressure to be lifted.

• **The question:** Is a hydrous or carbonated silicate melt lighter than about $3,864\ \mathrm{kg/m^3}$ along the ascent?

• **What the model predicts:** The pressure test passes below $3,864\ \mathrm{kg/m^3}$; at $3,500\ \mathrm{kg/m^3}$ a margin of $10.49\ \mathrm{GPa}$ is left, while at $4,000\ \mathrm{kg/m^3}$ it does not pass (Paper 3).

• **Data:** Shock-compression and diamond-anvil-cell data on hydrous and carbonated melts; first-principles molecular dynamics.

• **Method:** Experiments and ab-initio molecular dynamics (quantum simulations at the atomic scale) at the relevant compositions, giving the density threshold as a function of volatile content.

• **Where in the series:** Paper 3, Further work item 3; §A.9–A.10; §D.1 row 1; Paper 1 §F.2 and §E.1; Paper 13 §D.3 (row R6).

## 5 The energy of the lift and the energy release of the parent

Lifting a lunar mass and a pear several times heavier out of the Earth's gravity takes a great deal of energy, which in the model is supplied by the descent of iron to the core, releasing gravitational energy. The topic asks whether the supply of energy matches the cost and at what rate it was delivered.

• **The question:** Does the evolving Earth release the work needed to lift the body and the pear, and at what rate?

• **What the model predicts:** Lifting the Moon costs of order $10^{30}\ \mathrm{J}$ and lifting the pear about $8 \times 10^{30}\ \mathrm{J}$ (5.4 times the Moon's), while the descent of the iron frees of order $10^{31}\ \mathrm{J}$ (Paper 3).

• **Data:** Parameters of core formation and of cooling under a steam atmosphere.

• **Method:** A thermal-evolution model of an Earth whose iron is descending, contracting and cooling, with the duration of the feed derived from its output.

• **Where in the series:** Paper 3, Further work item 4; §A.12; §C.9; §D.1 row 3; Paper 1 §F.2.

## 6 The pressure test, repeated with the pear's mass and the flattened parent

A parent completing one rotation in $5.5\ \mathrm{hours}$ is flattened (characterised by the gravity coefficient $J_2$, which describes the equatorial bulge), so its gravity field differs from that of a sphere. Paper 3 estimated the effect of the flattening; what remains is to complete the pressure test with the pear's own mass included.

• **The question:** Does the source pressure still lift and hold the body and the pear in the field of a flattened parent?

• **What the model predicts:** With $J_2 \simeq 0.036$ the inward pull rises by $3\%$ ($5.4\%$ at the equatorial surface), and the cost of the lift rises by about $4\ \mathrm{GPa}$ (within the margin of $10.49\ \mathrm{GPa}$); the test has still to be rerun with the pear's own mass (Paper 3).

• **Data:** The geometry and mass of the pear (Paper 3 §C.7).

• **Method:** Running the joint test of Paper 3 §A.9–A.13 in a non-spherical gravity field.

• **Where in the series:** Paper 3 §C.9; §A.13.

## 7 The construction rate: feed flux over the window

The flow of a viscous melt through a passage grows as the fourth power of the passage radius ($b^4$), so a broad passage is required to deliver the lunar mass in the permitted time without freezing and closing.

• **The question:** Does the net feed deliver the internal mass in the time available (about $400 + 100 + 100\ \mathrm{Myr}$, with $100\ \mathrm{Myr}$ as a sensitivity case)?

• **What the model predicts:** The flux is sensitive to aperture as $b^4$; a passage of about $1,000\ \mathrm{km}$ radius is required ($0.46$ lunar masses; a lateral load of order $10^{22}\ \mathrm{N}$) (Paper 3).

• **Data:** Viscosities of the melt; radii of the passage.

• **Method:** A model of pipe flow and freezing over the permitted ranges of viscosity and aperture.

• **Where in the series:** Paper 3 §D.1 row 2; §A.11; §A.13; Paper 1 §E.1.

## 8 The pear's self-holding equilibrium figure and its budget of angular momentum (R9)

The pear is the Earth's own material standing out toward the corotation line and holding itself up as a dynamic equilibrium figure (requirement R9 of Paper 13). A detailed estimate would replace the present approximation based on round cross-sections of uniform density and would close the budget of angular momentum.

• **The question:** Does a rotating, internally pressurised Earth admit an equilibrium figure with a body at its tip — a base of about $9,000\ \mathrm{km}$ narrowing in one arc to a contact of $3,000\ \mathrm{km}$,

wholly inside the corotation line — and does that figure close the angular-momentum budget?

• **What the model predicts:** The narrowest self-holding base grows from $7,800$ to $8,800$ km (at $3,000$ kg/m$^3$) between the start and release; the pear's mass is $8.8 - 12.1$ lunar masses for bases of $9,000 - 11,000$ km; the pear passes to the Moon about a sixth of the system's angular momentum, which is taken from the Earth's spin and offset by the centralising core (Paper 3).

• **Data:** The parent state of the series (no observations are needed).

• **Method:** A calculation of an equilibrium figure with a free surface, including the pear's own gravity along its axis and the internal flow.

• **Where in the series:** Paper 3 §C.7, §C.8, §C.9; Paper 13 §C.6 (R9); Paper 1 §E.1 (R9, awaiting); Paper 2 §A.4.

## 9 The northward lean of the pear: oblique feed from a source north of the equator

The base of the pear lay near the equator ($1.9^\circ$N), whereas the Moon was released from latitude $22.5^\circ$N. Since rotation pushes a raised mass toward the equator and not away from it, the model attributes the northward lean to the oblique feed path from a deep source lying north of the equator.

• **The question:** Can a deep source north of the equator feed obliquely enough to take the tip from a base near $1.9^\circ$N to about $22.5^\circ$N, against the equatorward push of the rotation?

• **What the model predicts:** The lean comes from the feed path, not from the rotation (Paper 3).

• **Data:** The position of the base (Kiritimati) and of the low-velocity province (LLVP) beneath the Pacific, read with reservation as the returned pile.

• **Method:** A geometry of the deep source and of the feed path in a rotating, flattened parent.

• **Where in the series:** Paper 3 §D.1; §A.4; §B.1.

## 10 The Earth's core during the pear

A pear of about nine lunar masses on one side of the Earth moves the Earth's centre of mass toward it and draws the dense core after it; when the pear fell back, the core returned to its

place. The topic asks whether this episode left a signature in the east–west asymmetry of the solid inner core.

- **The question:** Was the Earth's core drawn $270 - 370\ \mathrm{km}$ toward the pear while the pear stood, and did it return after the pear fell?
- **What the model predicts:** A displacement of $270\ \mathrm{km}$ at the start (base $9,000\ \mathrm{km}$) growing to $370\ \mathrm{km}$ at release (base $11,000\ \mathrm{km}$), and a return after compensation; a hemispherical structure of the inner core aligned with the pear's side is the record to look for (Paper 3).
- **Data:** The east–west asymmetry of the inner core in seismic travel times; the age of the inner core (the test weakens if the inner core is much younger than the Moon).
- **Method:** A three-dimensional hydrostatic model of the Earth with the pear standing and after its fall, following the deficit left in the source.
- **Where in the series:** Paper 3 §E.5; Further work item 7.

## 11 The classical fission route, recomputed

The classical fission model (George Darwin onward) was rejected because it required far more angular momentum than the Earth–Moon system has, whereas the connected transfer of the LEF model needs almost exactly today's angular momentum (Paper 2). A modern computation would make the comparison exact.

- **The question:** What does a modern computation of the Jacobi sequence and its pear-shaped branch give for this mass and density?
- **What the model predicts:** The classical route needs $2.74$ times today's angular momentum at the Jacobi bifurcation point, against $1.007$ for the connected transfer (Paper 2).
- **Data:** Tabulated Jacobi and Poincaré sequences.
- **Method:** A numerical sequence of rotating fluid equilibria, with the pear-shaped point verified.
- **Where in the series:** Paper 2, Further work item 6; §A.6.5.

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PART B

# The body on the tip — shape, rounding, release

Part B covers the Moon while still attached to the tip of the pear: a foamed body fed through its base and wrapped in the Earth's outer shell, which passed through outward rounding, flattening and inward rounding while its contact narrowed from $6,000$ to $3,000$ km, until its release at the corotation line. Topics 12–23 examine the stability of the body, the rheology of the shell, the mechanics of detachment, and the structure of the Moon's crust and porosity.

**Topics of Part B**

| No. | Topic | Main tool | Readiness |
|---|---|---|---|
| 12 | 3-D simulation of the connected body | SPH / finite elements | New modelling |
| 13 | Stiff shell over a mobile base | Thermal–rheological balance | New modelling |
| 14 | From $a/c \approx 1.20$ to round | Elastic–viscous figure | New modelling |
| 15 | The drift of the release latitude | Viscous-layer integration | New modelling |
| 16 | Stefan adhesion and failure of the contact | Laboratory, numerical | New data |
| 17 | The trapped pressure at detachment | Momentum balance | New modelling |
| 18 | Survival inside the fluid Roche limit | Visco-elastic calculation | New modelling |
| 19 | The layered far-side crust | Central-peak compositions | Public data |
| 20 | The near-side crust over the contact | Thermal model, GRAIL | New modelling |
| 21 | KREEP as the torn edge of the sial (R12) | Th, K maps; anorthosites | Public data |
| 22 | High-pressure phases in contact-zone samples | Laboratory, Apollo 14 samples | New data |
| 23 | Whole-Moon porosity by depth | GRAIL porosity inversions | Public data |

## 12 A three-dimensional simulation of the connected body

The series has specified the initial conditions of the connected body; what remains is to run a full three-dimensional simulation that follows its stability and the order of its stages of evolution up to release.

- **The question:** Is a body of the lunar mass, foamed to $1,300$ kg/m$^3$ on a $6,000$ km tip and fed through its base at $150$ GPa, stable, and does it go through outward rounding, flattening

and inward rounding and release with the computed margin?

• **What the model predicts:** A stable dome of about $3,650\ \mathrm{km}$; the contact shrinks from $6,000$ to $3,000\ \mathrm{km}$; the order of the stages is well advanced before detachment (Paper 3).

• **Data:** None needed; the initial state is fully specified in the series.

• **Method:** A smoothed-particle (SPH) or finite-element simulation with a free boundary and one shared timing model for feed, cooling and failure.

• **Where in the series:** Paper 3, Further work item 2; §D.1; §A.5; Paper 1 §F.2, §E.3.

## 13 A stiff shell over a mobile base: the coupled thermal–rheological solution

The model requires at once a stiff outer shell that keeps the far-side craters during the ascent, and a soft interface at its base that lets it ride up on the lengthening dome.

• **The question:** Is there one rheological range in which the outer shell stays stiff enough to keep its craters while the interface beneath it stays mobile enough to ride up on the dome?

• **What the model predicts:** A shell about $18\ \mathrm{km}$ thick with viscosity $\eta \gtrsim 10^{23}\ \mathrm{Pa \cdot s}$ through the ascent, over a weak interface between sial and sima (Papers 3, 13); the making of the record, its preservation and the rise must overlap in time (Paper 6).

• **Data:** Viscosity laws for anorthositic and sialic crust; cooling under a steam atmosphere.

• **Method:** A coupled thermal and rheological balance along the construction.

• **Where in the series:** Paper 3 §D.1; §A.7; §D.4 item 4; Paper 13 §D.1, §D.3; Paper 6 §O.1; Paper 1 §E.1.

## 14 The step from $a/c \approx 1.20$ to round before release

Late in the attached stage (a day of $5.77\ \mathrm{hours}$ and a mean density of $2,600\ \mathrm{kg/m^3}$), a fluid body would be a flattened Maclaurin spheroid with its equatorial radius $a$ about $1.20$ times its polar radius $c$, whereas at release the Moon must be rigid and round. The calculation of the elastic–viscous step that carries the body from one state to the other is a research topic in itself.

• **The question:** How does the rigid shell carry the body from the Maclaurin ratio of about $1.20$ late in the attached stage ($5.77\ \mathrm{h}$, $2,600\ \mathrm{kg/m^3}$) to a round body at release?

• **What the model predicts:** At release the body is rigid and round: shear modulus $\mu \geq 10\ \mathrm{GPa}$, axis ratio $\leq 1.04$ (a requirement not computed in the series; Papers 2, 7).

• **Data:** None (a theoretical calculation).

• **Method:** An elastic–viscous figure calculation of a shelled body under decreasing spin and increasing density.

• **Where in the series:** Paper 7 §C.1 and front matter; Paper 2 §B.7; Paper 1 §C.7.

## 15 The drift of the release latitude below $22.5^\circ$

The latitude at which the Moon was released sets its first orbital inclination ($i_0$). While the contact shrank from $6,000$ to $3,000\ \mathrm{km}$ the body sat on a viscous layer, and the rotation pushed it gently south toward the equator; the size of this slip has not yet been computed.

• **The question:** How far did the body slip toward the equator on the viscous contact layer during the shrinking from $6,000$ to $3,000\ \mathrm{km}$?

• **What the model predicts:** Anchoring holds the body on the broad tip; in outward rounding it has freedom but little time; the drift can only lower the latitude, so $22.5^\circ$ is an upper bound on $i_0$ (with $25^\circ$ as a sensitivity case; its size is not computed in the series; Papers 3, 4).

• **Data:** The layer thickness $d_0$ ($20 - 50\ \mathrm{km}$), its viscosity, and the duration of stage 3.

• **Method:** An integration over the shrinking of the tangential term $\omega^2 r \sin\varphi\cos\varphi$ (the equatorward push of the rotation at latitude $\varphi$), resisted by the viscous layer.

• **Where in the series:** Paper 3 §C.5; Further work item 1; Paper 4, Part E; Paper 1 §B.7.

## 16 Stefan adhesion and the time of failure of the contact

The Moon’s contact with the pear was a thin, partly molten layer in which Stefan adhesion acted (resistance to the flow of fluid into the widening gap). The way this layer finally parts sets the release margin beyond the corotation line.

• **The question:** How does a partly molten layer between two bodies let go — by cavitation, by necking, by a passage from viscous to brittle failure — and when?

• **What the model predicts:** A slow stage, followed by an unrestrained burst when the gap reaches $d = 3d_0$; the release margin is $\Delta r \simeq 2d_0 \approx 210\ \mathrm{km}$ beyond the corotation line ($d_0 \approx 105\ \mathrm{km}$), where the search for the release distance in Paper 3 §C.5a finds the present

inclination and the present eccentricity meeting, with slightly more than half of the Moon across the line.

• **Data:** Laboratory separation of plates joined by partial melt.

• **Method:** Laboratory and numerical work on non-Newtonian Stefan adhesion, giving the range of the margin and the time of release.

• **Where in the series:** Paper 3, Further work item 5; §C.5; §C.5a; §D.1; Paper 1 §F.2 and §D.1 (the orange row of Paper 3).

## 17 The trapped pressure at detachment

Paper 3 lists three agents acting at separation, one of which is the pressure held beneath the body at the moment it let go; Paper 4 computes the release velocity without this term, and its contribution remains to be examined.

• **The question:** What part of the Moon's initial velocity came from the pressure held beneath the body when it let go?

• **What the model predicts:** The trapped pressure is one of the three agents acting at separation in Paper 3, and is not included in the release velocity of Paper 4 (Paper 7).

• **Data:** None (a theoretical calculation).

• **Method:** A momentum balance of the release including the trapped pressure, to check whether the release velocity of Paper 4 needs correcting.

• **Where in the series:** Paper 7, Further work item 5; §C.6; Paper 3 §C.7–C.8; Paper 1 §F.2.

## 18 Survival inside the fluid Roche limit: the deformation solution

The young Moon was released at a distance of about $17,500\ \mathrm{km}$ ($17,496\ \mathrm{km}$, about $210\ \mathrm{km}$ beyond the corotation line; Paper 3, §C.5a), outside the rigid Roche limit but inside the fluid one; a full visco-elastic calculation would replace the order-of-magnitude estimate of the deformation given in the series.

• **The question:** Does a rigid body of radius $1,890\ \mathrm{km}$ survive $10^{1} - 10^{3}$ years at a distance of about $17,500\ \mathrm{km}$ without harmful deformation, heating or mass loss?

• **What the model predicts:** The body lies about $7,200\ \mathrm{km}$ beyond the rigid Roche limit ($10,314\ \mathrm{km}$) and inside the fluid limit for $2,600\ \mathrm{kg/m^3}$ ($19,971\ \mathrm{km}$), and survives because it was already rigid (Papers 2, 4).

- **Data:** None (a theoretical calculation).
- **Method:** A visco-elastic calculation replacing the order-of-magnitude estimate of the deformation.
- **Where in the series:** Paper 2, Further work item 5; §D.1; Appendix E; Paper 4, Part E; Paper 1 §D.1 (the orange row of Paper 4).

## 19 The layered far-side crust

In the model the far side was built layer upon layer (sial gathered from the Earth's surface and returned material that landed on the rising shell), the later layers being lighter and richer in fluids; the central peaks of large craters bring up material from known depths and serve as natural drill cores.

- **The question:** Does the iron content of the far-side crust rise with depth, more strongly than on the near side?
- **What the model predicts:** The far side is sialic in the sense of the early Earth's surface and built in layers whose iron rises with depth, and its thickness does not depend on the local history of basins (Paper 3).
- **Data:** Central-peak compositions (Kaguya, Moon Mineralogy Mapper M3, LRO) with their depth of origin; Chang'e-6 material; GRAIL residuals after removal of the basins.
- **Method:** A comparison of peak composition against depth of origin (far side against near side) and the building of a construction model that predicts the steps.
- **Where in the series:** Paper 3, Further work item 6; §B.7; §D.1; §D.4 item 8; Paper 1 §F.3 item 3.

## 20 The near-side crust over the contact

As the body peeled away from the pear, the lower ring of the dome came away with its sial and became the marginal belt, while over the $3,000\ \mathrm{km}$ contact the heat of attachment melted the sial and smoothed it.

- **The question:** To what depth did the heat of the contact melt the sial, and does this give the crustal thickness over the contact against the thickness of the marginal belt?
- **What the model predicts:** The lower ring of the dome peeled away together with its sial (the marginal belt), and over the $3,000\ \mathrm{km}$ contact the sial was melted and smoothed (Papers 3, 13).

- **Data:** GRAIL crustal-thickness models.
- **Method:** A thermal model of the contact, compared with the measured crustal thickness.
- **Where in the series:** Paper 3, Further work item 8; §C.4a; Paper 13 §C.4.

## 21 KREEP as the torn edge of the sial (R12)

The model reads the KREEP component (potassium, rare-earth elements and phosphorus, concentrated around Oceanus Procellarum) as the torn edge of the terrestrial sial, folded inward at the rim of the base as the body peeled away; this requires that the most primitive anorthosites lost the potassium and thorium that went into it.

- **The question:** Does KREEP lie at the rim of the base, around Procellarum, and did the most primitive anorthosites lose the K and Th of the Earth's sial?
- **What the model predicts:** The torn edge of the sial was folded inward at the rim of the base as the body peeled away (Papers 3, 13).
- **Data:** Thorium and potassium maps of Lunar Prospector and Kaguya; anorthosite compositions.
- **Method:** Mapping the distribution of KREEP against the rim of the base and checking trace elements in primitive anorthosites.
- **Where in the series:** Paper 13 §C.6 (R12); Paper 3 §C.10; Paper 1 §E.1 (R12, awaiting).

## 22 High-pressure phases from slow pressure in samples from the contact zone

As the contact narrowed from $6,000$ to $3,000\ \mathrm{km}$, the load on the interface produced a high but slow pressure. Unlike an instantaneous shock wave, which turns feldspar into maskelynite glass, slow pressure allows recrystallisation into dense phases (coesite and stishovite from silica, jadeite from feldspar); the Apollo 14 landing site at Fra Mauro lies inside the contact dome (about $29^\circ$ from its centre).

- **The question:** Do samples from the contact dome — Apollo 14 at Fra Mauro first — contain high-pressure phases formed by slow pressure (coesite, jadeite, stishovite), and not only shock glass?
- **What the model predicts:** Samples from inside the contact dome carry crystalline high-pressure phases, while similar samples from the belt carry shock features only; their

absence from the dome, or equal presence in the dome and the belt, would not support the reading.

• **Data:** Existing Apollo 14 samples from the Fra Mauro formation ($3.6^\circ\mathrm{S}, 17.5^\circ\mathrm{W}$, NASA collection); as a control, highland samples from Apollo 16 and Apollo 17 (both about $51^\circ$ from the centre of the contact, in the belt).

• **Method:** Raman spectroscopy, electron microscopy and microprobe analysis of thin sections under criteria fixed in advance.

• **Where in the series:** Raised in the author's working notes (collection of 16.10.2026); the contact dome ($0^\circ - 31^\circ$ about $24.1^\circ\mathrm{N}, 335.2^\circ\mathrm{E}$) in Paper 3 §B.4 and Paper 13 §B.8; the narrowing of the contact in Paper 3 §C.4–C.5.

## 23 Whole-Moon porosity by depth: the record of the foam

The body began as a foam at a density of about $1,300\ \mathrm{kg/m^3}$, was compacted to $2,600\ \mathrm{kg/m^3}$ at release and to $3,344\ \mathrm{kg/m^3}$ today, so residual porosity (about $12\%$ on average in the crust) survives only in the upper kilometres, where lithostatic pressure has not closed the pores.

• **The question:** Does the porosity of the whole Moon, not only of South Pole–Aitken, fall with depth as a foam compacted under pressure should leave it, and at what depth do the pores close?

• **What the model predicts:** The foam collapsed at the depth where the pressure sufficed, and residual porosity survives only in the upper kilometres; the test that separates this from surface fracturing lies in the closure depth and its spatial distribution.

• **Data:** GRAIL grids of bulk density, grain density and porosity (Wieczorek et al. 2013), and the linear and exponential depth models of Besserer et al. (2014); for comparison, Paper 8 found $18.9\% - 20.3\%$ at the surface, and at $10\ \mathrm{km}$ found $6.2\% - 7.9\%$ in the far-side highlands and $9.6\% - 12.8\%$ in the rest of the far side.

• **Method:** Area-weighted means of porosity at the surface and at depth in the exposure zones and over the whole Moon (with identical masking of the maria and the large basins), and a comparison of the closure depth with a model of the foam compacting under its own weight.

• **Where in the series:** Paper 8 §B6; Paper 7 §C.3; Paper 2 Appendix G, row E42; the author's working notes (collection of 16.10.2026; parallel to topic 92 for Phobos and Deimos).

PART C

# The orbit

Part C covers the Moon after release. In the model it left about $210\ \mathrm{km}$ beyond the corotation line (Paper 3, §C.5a), at a distance of about $17,500\ \mathrm{km}$ from the Earth's centre, on a nearly circular orbit inclined at the latitude of its release ($22.5^\circ$), and tidal forces carried it outward to a distance of $384,400\ \mathrm{km}$. Topics 24–29 examine the orbital and thermal evolution, the dissipation in the Earth, the Moon's fossil figure and the reconstruction of the early latitude.

### Topics of Part C

| No. | Topic | Main tool | Readiness |
|---|---|---|---|
| 24 | Coupled thermal–orbital integration | Secular integrator | New modelling |
| 25 | The Earth's dissipation history | Ocean-resonance models | New modelling |
| 26 | The onset of recession | Tidal integration | New modelling |
| 27 | The magma-ocean switch for a one-sided lid | Asymmetric thermal model | New modelling |
| 28 | The fossil figure | Closure geometry, GRAIL | New modelling |
| 29 | The early latitude, reconstructed | Structural reconstruction | New modelling |

## 24 The coupled thermal–orbital integration from the release state

Paper 4 follows the Moon from release to the present with a reduced law and reaches the measured inclination of $5.145^\circ$; the full test requires a coupled integration that follows the tides in both bodies as they cool, through the Laplace-plane transition and the Cassini state.

- **The question:** Does the release state ($r_0 = 17,496\ \mathrm{km}$, $a_0 \approx 18,163\ \mathrm{km}$, $e_0 \simeq 0.037$, $i_0 = 22.5^\circ$, a day of $5.77\ \mathrm{hours}$, $J_2$ scaled with $\omega^2$) evolve to today's inclination ($5.145^\circ$), today's eccentricity and a distance of $384,400\ \mathrm{km}$ within $3.873\ \mathrm{Gyr}$?
- **What the model predicts:** The reduced law gives $5.145^\circ$ at today's distance ($5.788^\circ$ from $25^\circ$); the full test passes through the Laplace-plane transition and the Cassini state (Paper 4).
- **Data:** Lunar laser ranging (LLR); the present orbital elements.

• **Method:** A secular integrator with time-dependent dissipation in both bodies, together with a short Newtonian integration that checks the initial ellipse with each separation velocity and in the flattened field.

• **Where in the series:** Paper 4, Further work item 1; Part E; §D.1; Paper 1 §F.2 and §D.1 (the orange rows of Paper 4).

## 25 The Earth's dissipation history

The strength of the tidal push that carries the Moon away depends on the Earth's response ($k_2/Q$), and no single constant $Q$ fits both the present rate of recession and the age of the Moon.

• **The question:** What rate function $k_2/Q(t)$, starting from a day of $5.77\ \mathrm{hours}$, fits both today's recession ($3.8\ \mathrm{cm/yr}$) and the age since release?

• **What the model predicts:** No single constant $Q$ fits both: $Q \simeq 12$ gives the present rate, while $Q \simeq 29$ gives $3.87\ \mathrm{Gyr}$ (Paper 4).

• **Data:** Tidal rhythmites (layered sediments that record ancient tides and lengths of day); past length-of-day records; the geometry of the ocean basins.

• **Method:** Ocean-resonance models of the early Earth.

• **Where in the series:** Paper 4, Further work item 2; §D.2–D.3; Paper 2, Further work item 2; Paper 1 §D.1.

## 26 The onset of recession

The Moon was released tens of kilometres beyond the corotation line, and the model requires that tidal recession begin at once, from the first orbit.

• **The question:** At what rate does the Moon begin to recede from a release made a few tens of kilometres beyond the corotation line?

• **What the model predicts:** $\langle \dot{a} \rangle > 0$ from the first orbit, where $\langle \dot{a} \rangle$ is the mean rate of change of the orbital radius (Papers 2, 4).

• **Data:** None (a theoretical calculation).

• **Method:** A tidal integration with frequency-dependent rheology in both bodies, starting from the release state of Paper 3.

• **Where in the series:** Paper 2, Further work item 4; §B.8; Paper 4 §B.2.

## 27 The magma-ocean switch for a one-sided lid

The dissipation of tidal energy in a partly molten Moon differs from that in a solid one, and the survival of the orbital inclination depends on when the passage from molten to solid took place for a Moon with a thick lid on the far side and a thin one on the near side.

- **The question:** When did a Moon with a thick far-side lid and a thin near-side lid complete its solidification, and was it before $20$ Earth radii?
- **What the model predicts:** The survival of the orbital inclination depends on the timing of this solidification switch (Paper 4).
- **Data:** Symmetric models of the lunar magma ocean published in the literature.
- **Method:** Building a thermal model of a Moon with an asymmetric lid.
- **Where in the series:** Paper 4, Further work item 4; §D.2.

## 28 The fossil figure

The Moon's fossil departures of figure — the polar flattening coefficient $J_2$ and the coefficient of elongation toward the Earth $C_{22}$ — were set in the model when the dome closed, and not by late tidal locking.

- **The question:** Does the figure left by the closure of the dome match the Moon's present degree-2 figure, and was the synchronous rotation kept from the attached stage?
- **What the model predicts:** The body carries its figure from the closure and not from late tidal locking, and has kept one face toward the Earth since the attached stage (Paper 4; Paper 13, R4).
- **Data:** GRAIL degree-2 gravity; LOLA topography.
- **Method:** Deriving the expected values of $J_2$ and $C_{22}$ from the closure geometry of Paper 3 and comparing them with the present values.
- **Where in the series:** Paper 4, Further work item 5; §C.3; Paper 13 §A.4, §B.4 (R4); Paper 11, Part M item 4.

## 29 The geographic link: the early latitude, reconstructed independently

An initial inclination of $i_0 = |\lambda| = 22.5^\circ$ is derived from the centre of the contact on the Earth; an independent reconstruction from the structure of the Pacific and the Moon's rotation,

without fitting to the orbit, would confirm this link.

- **The question:** Can the latitude of the Moon's centre of mass at release be reconstructed from the structure and the rotation alone, without fitting to the orbit?
- **What the model predicts:** A latitude of $22.5^\circ$ (the centre of the $3,000\ \mathrm{km}$ contact), with the identity $i_0 = |\lambda|$ holding independently of the reconstruction (Paper 4).
- **Data:** The structure of the Pacific (topics 80–90); the centre of the contact on the Moon.
- **Method:** A structural and rotational reconstruction.
- **Where in the series:** Paper 4, Part E; Paper 1 §D.1 (the awaiting row of Paper 4).

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PART D

# The maria and the openings

In the LEF model a lunar mare is not an impact basin later flooded by lava, but a structural opening above a feed dome — a site where hot, volatile-rich melt rose from below through the shell of the body that sat on the Earth's pear and poured out as basalt (Paper 3, Paper 5). The openings were exposed in a fixed order as the front of the retreating cover passed over them, and each sealed when the melt supply that kept it open failed. Topics 30–43 test this mechanism mare by mare (13 of them can be done at once on public data, and only the first needs new modelling).

### Topics of Part D

| No. | Topic | Readiness |
|---|---|---|
| 30 | The engine of the openings: critical flux and times of sealing | New modelling |
| 31 | The order of sealing | Public data |
| 32 | Thorium at high resolution over the feed domes | Public data |
| 33 | The rim ring of the volatile-rich maria | Public data |
| 34 | Gradual exposure within one mare | Public data |
| 35 | Vents and conduits: the character of an opening by its zone | Public data |
| 36 | The titanium part of Study 13 | Public data |

| | | |
|---|---|---|
| 37 | The collar of the open far maria | Public data |
| 38 | The pits and caves of the Moon: vapour voids, not lava tubes | Public data |
| 39 | The crust jump under an opening, by the age of its basin | Public data |
| 40 | The diameter of an opening by the duration of its feeding | Public data |
| 41 | The rim of a mare as a layered record: rim height and the duration of ejection | Public data |
| 42 | Blunt or sharp: the crest where a large crater's rim is missing on the side of the plain | Public data |
| 43 | Gravity traces of the craters that vanished into the active maria | Public data |

## 30 The engine of the openings: critical flux and times of sealing

An opening stays open as long as the melt passing through it brings in heat at a rate that balances the heat lost to the walls of the cold shell (equation 4 of Paper 5); when the flow weakens below the critical flux, the conduit freezes and seals.

- **The question:** For each mare, did the opening stay alive until the retreating front exposed it?
- **What the model predicts:** An opening seals when the melt flux falls below the critical flux, at which the heat carried in balances the heat lost to cooling (Paper 5, equation 4), and the openings outlived the moment of their exposure.
- **Data:** Areas, depths and ages of the maria; the exposure generations of Paper 5, §I.4.
- **Method:** Computing the critical flux and the time of sealing for each mare and comparing them with the time of its exposure by the front.
- **Where in the series:** Paper 5, Further work item 1, §§H.3, I.6, M.3; Paper 1, §F.2.

## 31 The order of sealing

Since the openings were fed from the contact (the region that last touched the terrestrial column), the openings nearest the contact lost their supply last, so their youngest basalts are younger than those of more distant maria (with the concentration of radiogenic, heat-producing thorium in the PKT region held fixed).

- **The question:** Does the youngest basalt of each mare become younger toward the centre of the contact, with thorium held constant?
- **What the model predicts:** The maria of the contact sealed last (Paper 5).

• **Data:** Crater-count and radiometric ages of the youngest units in each mare (such as the ages of Hiesinger and colleagues); thorium maps.

• **Method:** A regression of the youngest age on the angular distance from the centre of the contact fixed in advance, with thorium as a control variable.

• **Where in the series:** Paper 5, Further work item 6, §§K.2, M.3; Paper 1, §F.3, item 8.

## 32 Thorium at high resolution over the feed domes

Paper 5 requires a thorium minimum exactly over the channels of the feed domes, because the melt that rose through them arrived beneath the thorium-bearing layer and diluted it.

• **The question:** Is thorium lowest exactly over the channels of the feed domes?

• **What the model predicts:** The minimum lies over the channels, as Paper 5, §J.6 requires.

• **Data:** The reconstructed (deconvolved) thorium map at a resolution of $26\ \mathrm{km}$.

• **Method:** Comparing the thorium concentration over each dome with the rings around it at high resolution.

• **Where in the series:** Paper 5, Further work item 5, §J.6; Paper 1, §F.3, item 8.

## 33 The rim ring of the volatile-rich maria

The margins of the maria were built from material sprayed out while the opening was active; whereas Paper 5 ruled out a ring of small craters on all rims taken together, the narrower and stronger prediction is that such a ring marks the maria richest in the volatiles that drove the spraying.

• **The question:** Do the maria richest in volatiles carry a ring of small craters on their rims?

• **What the model predicts:** Not every rim carries a ring (Paper 5, §M.1); the ring belongs to the volatile-rich maria, whose spraying built the margins (Paper 5, Paper 6).

• **Data:** Catalogues of small craters; maps of pyroclastic deposits and dark mantle deposits; hydration maps from the Moon Mineralogy Mapper ($M^3$).

• **Method:** Comparing the density of small craters on the mapped rim against the outer terrain (with the maria removed on both sides) against the measured volatile content of each mare.

• **Where in the series:** Paper 5, Further work item 2, §§F.3, H.2, M.1, M.3; Paper 6, §O.1.

## 34 Gradual exposure within one mare

Since the exposure front moved along the body over time, a large mare that this front crossed was exposed gradually from one side to the other, so the side exposed first carries a longer crater record.

- **The question:** Does a large mare that the exposure front crossed differ from one side to the other, in the direction of exposure?
- **What the model predicts:** The mare was exposed side after side over time (Paper 5).
- **Data:** Crater preservation, fill and structure across large maria.
- **Method:** Analysis of profiles across the mare along the direction of exposure.
- **Where in the series:** Paper 5, §§I.3, M.3.

## 35 Vents and conduits: the character of an opening by its zone

The model distinguishes exposed openings that ejected material (crown and flank) from covered openings that only conducted melt until release (beneath the base), a difference expressed in the signature of the mascons in the gravity gradient.

- **The question:** Do openings that ejected while exposed (crown and flank) differ measurably from openings that only conducted under cover (beneath the base)?
- **What the model predicts:** In the belt, mascons with rings; in the contact, mascons without rings; beyond it, basins on thick crust (Papers 3, 5, 13).
- **Data:** Secondary-crater fields; crustal thickness beneath each basin; composition of the earliest fill; GRAIL gravity gradients; a rim-height index.
- **Method:** Grouping the basins in advance into a complete list with height indices and comparing them with models of impact, fill, subsidence and intrusion.
- **Where in the series:** Paper 3, Further work item 9, §B.5; Paper 5, §§D.6, M.3; Paper 13, Further work item 6, §§C.3, D.3 (related to topic 122, tests T3 and T4).

## 36 The titanium part of Study 13

Study 13 of Paper 5 asked whether the melt of the contact (which rose from greater depth) diluted thorium relative to titanium more than the melt beyond it; while the thorium part passed the test, the titanium part did not separate the zones (Imbrium does not separate, and Copernicus H shows the opposite sign).

• **The question:** Is the thorium-to-titanium ratio lowered more inside the contact than beyond it?

• **What the model predicts:** Dilution is expected, but the existing data did not separate the zones (Imbrium does not separate; Copernicus H shows the opposite; the thorium part held; Paper 5).

• **Data:** Thorium and $\mathrm{TiO_2}$ maps.

• **Method:** A reading of titanium at higher resolution (LROC Wide Angle Camera, $M^3$), with the maria and the plains separated within each zone.

• **Where in the series:** Paper 5, §§J.4, M.2 (Study 13, orange); Paper 1, §D.1.

## 37 The collar of the open far maria

Paper 5 found a raised collar of returned material around the far-side maria taken together (and most clearly around the buried ones among them), but around the ten open far-side maria alone (median collar $207\ \mathrm{m}$) it has not yet been shown to stand above the control.

• **The question:** Does the raised collar stand above the control around the open far maria alone?

• **What the model predicts:** The margins were built from layers of the mare's own returned material; a collar is found around the far maria together and clearly around the buried ones, but not yet around the 10 open far maria (median $207\ \mathrm{m}$) (Paper 5).

• **Data:** LOLA and SLDEM topography (the LRO laser altimeter merged with the Kaguya stereo terrain model).

• **Method:** Ring profiles with a larger or better-matched control.

• **Where in the series:** Paper 5, §§F.3, M.2 (Study 16, orange); Paper 1, §D.1.

## 38 The pits and caves of the Moon: vapour voids, not lava tubes

Unlike the literature, which sees the pits of the maria as skylights into shallow lava tubes, the author proposes that the pits and caves (including the passage found by radar beneath the pit of Mare Tranquillitatis) are deep, branching voids left by trapped vapour that broke out.

• **The question:** Are the pits and caves of the Moon voids left by trapped vapour that broke out, and not lava tubes?

• **What the model predicts:** Vapour voids (deep, scattered networks not confined to the area where lava flowed); the deciding case is a cave on the far side far from any basalt, which cannot be a lava tube.

• **Data:** The LROC catalogue of lunar pits (Wagner and Robinson 2014 and its updates); the LROC mare map; LROC Narrow Angle Camera images of the pit walls; Mini-RF radar (Carrer and colleagues 2024).

• **Method:** Mapping each pit by its setting and its distance from the nearest mapped basalt (with the far-side pits far from basalt listed apart), and examining the geometry of the void in images and radar (long and shallow against deep and branching).

• **Where in the series:** Raised in the author's working notes (collection of 16.10.2026); related to Paper 5, §I.5 and Paper 8, §B6.

## 39 The crust jump under an opening, by the age of its basin

Paper 5 measured the "crust jump" (the thinning that the feed dome produces beneath the centre of the opening relative to the ring around it) and found it at 29 of the 36 openings (median $17\ \mathrm{km}$ against $1\ \mathrm{km}$ at ordinary craters). The topic adds the dependence on age: an old basin formed on a hot Moon underwent viscous relaxation and lost part of its jump, while a young basin kept it (the literature reads the subdued gravity of some old basins in the same way, e.g. Kamata and colleagues 2013).

• **The question:** Is the crust jump under an opening, relative to its size, larger at young basins and smaller at old ones?

• **What the model predicts:** Yes; at an old basin formed while the Moon was hot the jump relaxed, while a young basin kept more of it.

• **Data:** The crust jumps of Paper 5, §D.2 (mean of the four GRAIL models: centre to $0.3R$ against a ring at $1.3-2.0R$); the relative order of the basins by stratigraphic overlap (Wilhelms 1987) and by the density of superposed craters (Fassett and colleagues 2012), without conversion to years.

• **Method:** A rank correlation between the jump normalised to the radius (or between basins of similar size) and the relative order by overlap and by crater density.

• **Where in the series:** Paper 5, §D.2 (Study 3); the dependence on age was raised in the author's working notes (collection of 16.10.2026).

## 40 The diameter of an opening by the duration of its feeding

Paper 5 predicts that the diameter of an opening is a function of the history of feeding, pressure, temperature, flux, viscosity and exposure: $D = F(\text{history of feeding}, P, T, Q, \eta)$, and divides the openings into three generations (generation I on the far side, generation II in the belt, generation III in the contact). The test of the full order of diameters by generation and by distance from the centre of the contact has not yet been made.

• **The question:** Are the openings fed longest under cover wider than those exposed and cooled early?

• **What the model predicts:** The diameter of an opening is ordered by the duration of its feeding (Paper 5, §I.6): the openings fed longest under cover are the widest (the six largest domes, $450 - 909\ \text{km}$, all lie on the near side, median $375\ \text{km}$ against $227\ \text{km}$ on the far side).

• **Data:** The diameters, zones and generations of the feed domes and maria in Paper 5, Appendix A; the GRAIL gravity field; the angular distance of each opening from the centre of the contact (Paper 5, §I.1).

• **Method:** Examining diameter against generation and against distance from the centre of the contact within the near side and across both sides (with buried and open domes measured in the same way).

• **Where in the series:** Paper 5, §§I.4, I.6; related to topic 30.

## 41 The rim of a mare as a layered record: rim height and the duration of ejection

In the model the margins of a mare were built from outside, layer upon layer, from the material the mare itself ejected (Paper 5, §F.3), so such a rim must be layered with a spread of ages, and its height grows with the duration of ejection (unlike a single collapse rim of one age).

• **The question:** Are the rims of the maria built of layers whose ages are spread over time, and does the rim height grow with the length of time over which its mare ejected material?

• **What the model predicts:** The higher the rim, the longer the ejection lasted, and the rim is layered with a spread of ages (Paper 5 found that 16 of 16 large floors lie below their rim, by a median of $3.4\ \text{km}$).

• **Data:** Rim heights from LOLA and SLDEM2015 (Paper 5, Study 6); crater-count ages of the rim units; LROC Narrow Angle Camera images and $M^3$ spectra; the order of exposure and the generation of each opening (Paper 5, §§I.1, I.4).

• **Method:** Comparing rim height against the duration of ejection (from the order of exposure and the times of sealing of topic 30), and searching for layering in the walls of massifs and for a spread of ages among the units of one rim.

• **Where in the series:** Paper 5, §F.3 (Study 16), §F.1; related to topics 33 and 37.

## 42 Blunt or sharp: the crest where a large crater's rim is missing on the side of the plain

On the plains of the contact, large craters have lost their rims on the side facing the plain; whereas the literature reads a low rim on a mare plain as flooding by basalt (which leaves a sharp upper crest), the model reads it as thermal erosion by the hot contact plain (which produces a blunt crest). Study 15 in Papers 5 and 6 found that on the plains of the contact the rim is missing 24% more often on the side facing the plain and the crest is blunter (its sharpness lower by 35 m/km², its width 50 m below the top greater by 0.14 km, and its angle wider by 5.3°), but only four matched pairs could be formed.

• **The question:** In a matched sample of craters of equal interior depth, are the crests on the plains of the contact blunter than on the plains beyond it?

• **What the model predicts:** Blunt crests on the plains of the contact (where the plain was hot), and sharp crests beyond it (where only flooding acted; Paper 5, §M.3).

• **Data:** SLDEM2015 at a resolution of 512 pixels per degree (60 m); finer stereo terrain models from the LROC Narrow Angle Camera; the crater catalogue of Wang & Wu; the LROC mare map.

• **Method:** Measuring radial profiles of the crest (curvature at the top, width 50 m below the top, and the angle between the inner and outer slopes) on a larger sample matched in diameter, latitude and interior depth between the plains of the contact and the plains beyond it.

• **Where in the series:** Paper 5, §§G.6, M.3 (Study 15); Paper 6, §E.6.

## 43 Gravity traces of the craters that vanished into the active maria

On the floors of the contact maria (703,000 km²) not one crater ≥ 20 km remains, whether melted or sharp (Paper 6, §D.2), because craters that landed there while the maria were active were swallowed by them and vanished; this is unlike craters buried in basalt (Evans and colleagues 2016), where a body with steep gravity margins remains.

• **The question:** Did the craters that vanished into the active maria of the contact leave a mark in the gravity field?

• **What the model predicts:** No crater-shaped body beneath the floors of the contact maria (no ring of steep gravity at a crater's margin, except where the crater's margin is visible above the surface), and at most a shapeless mass excess, as beneath the melted craters (Paper 6, §D.3).

• **Data:** The GRAIL crustal model and the Bouguer field; gravity fields of higher degree; the 34 maria and the cores of their feed domes (Paper 5, Appendix A); the catalogue of buried craters of Evans and colleagues (2016).

• **Method:** Searching for crater-sized rings in the Bouguer field and its gradients on the floors of the contact maria against random points (to remove the effect of the lower dome, as in Study 14), and checking the steepness of the margins of the rings of Evans and colleagues by zone.

• **Where in the series:** Paper 6, §D.2, §D.3 (Study 14).

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PART E

# The craters and their source

In the LEF model about 99% of the lunar craters were made by returned terrestrial material — material thrown from the openings of the Earth and of the body itself, which landed on the shell at about the speed at which it left (Paper 6). This part collects the 14 topics (44–57) that follow the physical chain from the exit velocity at the opening, through the passage of the steam atmosphere and the shock at landing, to the reading of the chemical composition and the spatial distribution of the craters (including topic 49, one of the two deciding key observations of the model).

### Topics of Part E

| No. | Topic | Readiness |
|---|---|---|
| 44 | The exit velocity from the openings | New modelling |
| 45 | Passage through the steam atmosphere | New modelling |
| 46 | Shock in a hot, layered target; scaling at g = 8.25 m/s² | New modelling |

| | | |
|---|---|---|
| 47 | The trend in a series of events | New data |
| 48 | The crater size distribution, propagated | New modelling |
| 49 | Seeing the returning body, not the ground | New data |
| 50 | Captured ejecta after release | New modelling |
| 51 | The longitude record: landing strength or maria? | Public data |
| 52 | The trailing-side excess: the yield of craters near 90°E | New modelling |
| 53 | The Earth as the last scatterer | Public data |
| 54 | Material after release poorer in iron | Public data |
| 55 | Iron of the high and low floors | Public data |
| 56 | Crater-count ages against direct ages | Public data |
| 57 | The rays of young craters: material within the surface | Public data |

## 44 The exit velocity from the openings

Paper 6 draws the energy of ejection from the volatiles (water and carbon dioxide dissolved in the melt at depth, which come out of solution as the pressure falls), but has not yet shown that the source pressures of Paper 3 produce the exit velocity needed for landing at 2 - 8 km/s.

- **The question:** Does explosive decompression of a volatile-rich melt, at the source pressures of Paper 3, give exit speeds such that the material returns at 2 - 8 km/s?
- **What the model predicts:** The energy of ejection comes from the volatiles; the speed of the returning material is a requirement not proved in Paper 6 (Paper 6, Paper 13).
- **Data:** Volatile contents (requirement R6); source pressures (Paper 3).
- **Method:** A model of explosive decompression giving the distribution of exit speeds.
- **Where in the series:** Paper 6, Further work item 2, §§J.1, O.1; Paper 13, §D.1.

## 45 Passage through the steam atmosphere

The ejected material crossed a dense steam atmosphere with a base pressure of 10 - 1,000 bar (Paper 6 gives a single example of a body 1 km across at 100 bar that keeps half its speed); the range of sizes that arrive intact has not yet been computed.

- **The question:** Which returning bodies of 0.1 - 10 km reach the shell intact through an atmosphere of 10 - 1,000 bar?
- **What the model predicts:** A compact returning body passes through and meets the shell (Paper 6).
- **Data:** None needed (a theoretical calculation).
- **Method:** A calculation of coupled gravity-and-drag trajectories, including heating and break-up.
- **Where in the series:** Paper 6, Further work item 3, §J.3.

## 46 Shock in a hot, layered target, and crater scaling at g = 8.25 m/s²

Unlike the basalt-on-basalt estimate of Paper 6, the model's target is a cold sialic shell over a hot, mobile base on a body whose surface gravity is 8.25 m/s² (five times the Moon's gravity today).

- **The question:** What shock pressure and what crater size does a dense returning body produce on a cold sialic shell over a mobile base?
- **What the model predicts:** This calculation replaces the basalt-on-basalt estimate of Paper 6.
- **Data:** Hugoniot data for sial and basalt.
- **Method:** Impedance matching (solving for the common pressure and the common particle velocity at contact) and crater-scaling laws for the gravity of the shell.
- **Where in the series:** Paper 6, Further work item 4, §J.2.

## 47 The trend in a series of events

Equation (14) of Paper 6 ties the shock in a returning body to the height of the target on the body, with the trend depending on the source of the material, the spin and the gravity field of the series of events.

- **The question:** Do the shock indicators change with the height of the target as equation (14) predicts, for one defined source, spin and field?
- **What the model predicts:** The trend is conditional on the path, as set in Paper 6, and is read within a series of events that share one source, spin and field.
- **Data:** Shock measurements in units whose age and origin are identified.

- **Method:** Controls on material, angle and position within an event.
- **Where in the series:** Paper 6, §§O.1 (row 2), K.3.

## 48 The crater size distribution, propagated forward

In the model the size–frequency distribution of the craters follows from the sizes and speeds of the material ejected from the openings, after break-up in flight, landing, saturation and later burial; Paper 6 holds the distribution as a constraint and not as a full forward test.

- **The question:** What size–frequency distribution follows from the ejected sizes and speeds, after break-up, landing, saturation and burial?
- **What the model predicts:** For now a constraint, not a failed test (Paper 6).
- **Data:** Crater catalogues (Robbins; Head and Kadish; Wang and Wu).
- **Method:** A forward model from the source to the surviving craters, without assuming saturation or a constant flux.
- **Where in the series:** Paper 6, §P.2.

## 49 Seeing the returning body, not the ground

This topic is one of the two key observations of the model in Part L: while the ground around a mare carries the mare's iron (6.9 → 3.1 wt% FeO within 150 km against 2.3 wt% in the control, Paper 5), the crater itself looks like its surroundings because the landing threw aside the thin coating; the signature of the returning body must therefore be read in the melt and glass formed at landing inside the crater.

- **The question:** Can the share of returned material inside a crater be identified, and does it carry the iron and thorium signature of its source mare?
- **What the model predicts:** The ground around a mare carries the mare's iron (6.9 → 3.1 wt% FeO within 150 km; control 2.3), but the crater looks like its surroundings because the landing threw the coating aside; the signature of the returning body is carried in the melt (Paper 5, Paper 6).
- **Data:** Melt and glass deposits inside craters; deconvolved thorium; future samples.
- **Method:** Reading the interiors of craters zone by zone; comparing the compositions of the melt and glass with the candidate source mare; a mixing model of the returning body and the target.

- **Where in the series:** Paper 6, §§L.2, L.3, O.1; Paper 5, §§C.7–C.8 (related to topic 123).

## 50 Captured ejecta after release

After the Moon was released at a distance of a few Earth radii, the Earth went on throwing material, part of which was captured by the Moon and made the few young craters on the maria.

- **The question:** What fraction of the Earth's ejecta after release did a Moon at a few Earth radii capture, as a function of launch speed and direction?
- **What the model predicts:** The Earth went on throwing after release, and part of it reached the Moon (Paper 3, Paper 6).
- **Data:** The few young craters on the maria; their shock and size record.
- **Method:** An orbital-capture calculation, together with the record that separates this population from late bodies from outside.
- **Where in the series:** Paper 6, Further work item 1, §M.1; Paper 3, §C.11.

## 51 The longitude record: landing strength or maria?

In the model the landings of ejecta on the leading hemisphere (90°W) were head-on and hard, while those on the trailing hemisphere (90°E) were soft; Paper 6 finds that serrated rims are 1.14 - 1.25 times as common at 90°W, and the effect of the asymmetric maria remains to be isolated.

- **The question:** How much of the difference between 90°W and 90°E in rims, peaks and floors is landing strength, and how much is the maria?
- **What the model predicts:** Hard landing on the leading hemisphere (90°W) and soft landing on the trailing hemisphere (90°E); serrated rims are 1.14 - 1.25 times as common at 90°W, but some of the confidence intervals of the larger rims include zero (Paper 6).
- **Data:** The four domes of Paper 6, §N.1; the catalogue of lunar impact-melt flows (Neish and colleagues).
- **Method:** Manual checking of central peaks and rims, linkage to the catalogue of melt flows, and computation of the same indices by crater age.
- **Where in the series:** Paper 6, Further work item 5, §§F.1, I.1, Study 6 (orange); Paper 1, §D.1.

## 52 The trailing-side excess: the yield of craters from the openings near $90^\circ\mathrm{E}$

The trailing hemisphere of the Moon holds more large craters than the leading hemisphere ($1.96$ times before masking the maria and $1.45$ times after); the model attributes this to the body's own ejecta landing toward the trailing hemisphere and to the larger number of openings near $90^\circ\mathrm{E}$, a link that remains to be computed quantitatively.

- **The question:** Did the more numerous openings near $90^\circ\mathrm{E}$ make enough craters $\geq 20\ \mathrm{km}$ to account for the trailing-side excess ($1.96$ raw, $1.45$ masked)?
- **What the model predicts:** The body's own ejecta lands toward the trailing hemisphere; the link is a requirement, not yet a computed result (Paper 6, Paper 7).
- **Data:** Crater density as a function of distance from each mare.
- **Method:** Carrying the ejection history of each opening through the flight calculations of Paper 6 to a landing yield.
- **Where in the series:** Paper 7, Further work item 9, §§E.2, A.9; Paper 6, §K.4; Paper 1, §F.2.

## 53 The Earth as the last scatterer

The last small craters on the smooth floors of the maria are more numerous on the near side ($1.12 - 1.27$ times after removal of secondary craters, Paper 5 Study 9); the open step is to count these craters by distance from each candidate source, so as to isolate the Earth's contribution.

- **The question:** Do the last small craters on the smooth floors increase with distance from the contact, and does a near-side excess remain once this rise is held fixed?
- **What the model predicts:** The near-side excess of the last small craters is $1.12 - 1.27$ after removal of secondaries (Study 9, green); the open test is a count by distance from each candidate source (Paper 5, Paper 6).
- **Data:** Fresh craters of $1 - 2\ \mathrm{km}$ diameter; masks of rays and of secondary fields.
- **Method:** Counts on the floors of the maria by distance from each candidate source.
- **Where in the series:** Paper 5, Further work item 3, §§F.6, M.3; Paper 6, §§L.7, O.1.

## 54 Material after release poorer in iron

After release the heavy material sank down the pear toward the Earth, and a gas-driven flow accelerated the light, small fractions to an escape/arrival speed of $\geq 8.8\ \mathrm{km/s}$ toward the Moon; a crater made on a mare by such material should therefore throw out ejecta poorer in iron than the basalt around it.

- **The question:** Are the ejecta of craters formed on the maria after release poorer in iron than the surrounding mare, with the depth of excavation controlled?
- **What the model predicts:** What reached the Moon from the Earth after release was the lighter fraction; among the craters within the basalt the result stays within the scatter of the maps (Paper 6).
- **Data:** $\mathrm{FeO}$ maps at higher resolution.
- **Method:** Comparing the composition of the ejecta with the surrounding mare, with crater size and depth of excavation (which brings up different layers) controlled.
- **Where in the series:** Paper 6, §§L.6, O.1, Study 12 (orange); Paper 1, §D.1.

## 55 Iron of the high and low floors

A crater floor on a mare that froze high held its volatiles and its light material and so should be poorer in iron than a floor that subsided; Paper 6 found a difference that does not exceed chance ($-0.44$ against $-0.64\ \mathrm{wt\%\ FeO}$ relative to the basalt outside the crater).

- **The question:** Are floors that froze high poorer in iron than floors that sank back?
- **What the model predicts:** High floors held their volatiles and their lighter material; low floors are richer, but against the basalt just outside the crater the difference ($-0.44$ against $-0.64\ \mathrm{wt\%}$) does not exceed chance (Paper 6).
- **Data:** $\mathrm{FeO}$ maps; floor heights relative to the basalt outside.
- **Method:** An analysis on a larger sample or at higher resolution.
- **Where in the series:** Paper 6, §§G.3, O.1, Study 11 (orange); Paper 1, §D.1.

## 56 Crater-count ages against direct ages of the same surface

Since in the model nearly all craters were made by returned material whose rate changed with the activity of the openings (and not by a constant external flux, as in the Neukum calibration curve), converting crater counts to years overstates the age of a surface relative to a direct

radiometric age (as with Chang'e-6, where counts gave $2.98 - 1.54\ \mathrm{Gyr}$ against a laboratory age of $2.807 \pm 0.003\ \mathrm{Gyr}$).

- **The question:** Where the same surface has been dated both by crater counts and directly, is the direct age systematically younger than the crater-count age?
- **What the model predicts:** Yes; a count converted to years at a constant flux overstates the time of exposure (Paper 6, Part M), and a direct age reads the time at which the surface cooled and closed, not the age of the opening beneath it (Paper 5, §K.1).
- **Data:** Published crater-count model ages and radiometric ages of the same units: the Apollo and Luna sites, Chang'e-5 (about $2.0\ \mathrm{Gyr}$) and Chang'e-6 (Paper 5, §K.1; Paper 8, B9).
- **Method:** Pairing each dated sample with the crater-count age of the same unit only (focusing on surfaces outside the Apollo and Luna calibration set and using counts published before the sample was dated; the relative clocks are kept and only the conversion to years is tested).
- **Where in the series:** Paper 6, Part M and the row "The ages" in the comparison table; Paper 5, §§K.1–K.2; Paper 8, B9 and B15 (related to topics 31 and 48).

## 57 The rays of young craters: material within the surface

Unlike the usual reading, which takes the rays of young craters as ejecta lying on top of the ground, the author proposes that a ray is a record frozen within the ground below the surface when material met a cold, glassy floor (such as Copernicus, Kepler and Aristarchus, which landed on a cold floor, against Diophantus and Delisle, which landed on a warm floor and were left without rays).

- **The question:** Are the rays of young craters a record frozen within the ground rather than ejecta lying on it, and do they form only where the material landed on a floor that had already cooled?
- **What the model predicts:** The rays lie within the ground and not above it; on the mare floors and basalt plains of the near side, craters that landed after the floor cooled carry rays (Copernicus, Kepler, Aristarchus) and those that landed while the floor was warm lack them (Diophantus, Delisle), so that rayless small and medium craters outnumber rayed ones; Tycho in the highlands is the test case for the depth of the rays.

• **Data:** LROC images and the LROC mare map; maps of iron and of surface maturity from the Kaguya Multiband Imager; the lists of rayed craters of Grier et al. (2001) and of Werner and Medvedev (2010 — 1,615 craters, Paper 7, §§D.3, E.4); near-side crater catalogues for counting rayless craters.

• **Method:** Mapping the extent of each ray system and dividing it between mare floors and basalt plains on the one hand and highlands on the other, per unit area and by size; comparing counts of rayed and rayless craters on the near-side floors in the same size classes; examining the floors and ejecta of small fresh craters within the rays of Tycho (dark material beneath the ray points to a thin surface layer, while bright material continuing at depth points to a ray within the ground).

• **Where in the series:** Raised in the author's working notes (collection of 16.10.2026; brought for his reading on 17.10.2026); related to Paper 5 §F.5, Paper 6 §L.6, Paper 7 §D.3, and topics 50 and 58.

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PART F

# The poles

Paper 7 reads the crowding of old craters toward the Moon's poles (a pole-to-equator density ratio of $1.20 \pm 0.05$ in the clean count) as a sign of the rounding of the attached body: the craters were made while the body was flattened by its spin, and as it rounded inward the polar surfaces contracted while the craters kept their diameters and moved closer together. Topics 58–67 test this sequence (7 of them can be done at once on public data, 2 need new modelling and 1 needs a new catalogue).

### Topics of Part F

| No. | Topic | Readiness |
|---|---|---|
| 58 | The young craters of the poles | Public data |
| 59 | The figure with strength and two-layer relaxation | New modelling |
| 60 | The tide on a body held at its base | New modelling |
| 61 | The spacings between polar craters | Public data |

| | | |
|---|---|---|
| 62 | The marks of the shell | Public data |
| 63 | The innermost caps | New data |
| 64 | The shallower polar craters | Public data |
| 65 | The alignment of elongated craters at middle latitudes | Public data |
| 66 | The polar scale: the onset of central peaks | Public data |
| 67 | Crater ellipticity, and the poles against the marginal belt | Public data |

## 58 The young craters of the poles

This is the cleanest test of the polar reading: craters formed after the Moon became round must be spread evenly in latitude (a pole-to-equator ratio of about $1.0$), unlike the ratio of $1.20$ in the old population; the present samples ($0.92$ and $1.12 - 1.16$) are consistent but too small to decide.

- **The question:** In a dated sample of about 150 young craters in the caps (and about 300 in the band), is the pole-to-equator ratio about $1.0$ or the $1.2$ of the old population?
- **What the model predicts:** Craters formed after the Moon was round are uniform in latitude ($0.9 - 1.1$); the present samples give $0.92$ and $1.12 - 1.16$ (Paper 7).
- **Data:** Mazrouei et al. (2019), Table S1; the 1,615 rayed craters of Werner and Medvedev (2010); new dated samples.
- **Method:** A clean count by latitude on a larger, verified sample.
- **Where in the series:** Paper 7, §§D.3, E.3 (item 3), Further work item 2; Paper 1, §F.3 item 7, §D.1 (the orange row of Paper 7).

## 59 The figure with strength and two-layer relaxation

A fluid Maclaurin figure gives an upper bound on the polar crowding, whereas a stiff shell ($10^{23}\ \mathrm{Pa \cdot s}$) over an interior of viscosity $10^{21} - 10^{22}\ \mathrm{Pa \cdot s}$ reduces the flattening from below and sets the residual deformation.

- **The question:** How much of the fluid flattening did a body with a stiff shell take up, and how did the interior ($10^{21} - 10^{22}\ \mathrm{Pa \cdot s}$) and the shell ($10^{23}\ \mathrm{Pa \cdot s}$) share the contraction?
- **What the model predicts:** The fluid figure gives the upper bound; a stiff shell reduces the prediction from below, and the deformation in the shell is shortening at the poles and

fracturing at the equator (Paper 7).

• **Data:** The measured clean ratio, $1.20 \pm 0.05$.

• **Method:** A calculation of a Maclaurin figure with strength and two-layer visco-elastic relaxation through the attached stage.

• **Where in the series:** Paper 7, Further work items 1 and 3, §§C.1–C.2, C.7; Paper 1, §F.2.

## 60 The tide on a body held at its base

While attached to the pear, the body was stretched along the line to the Earth by the tide and flattened by its spin; Paper 7 measures the order of the six axes, the dense crown of the far side and the ratio of the largest craters ($1.84 \pm 0.38$), and the tidal figure that would produce these numbers remains to be computed.

• **The question:** Does the tidal figure of a body held at its base give the order of the six axes as a number, the dense crown of the far side, and the ratio of the largest craters ($1.84 \pm 0.38$)?

• **What the model predicts:** The body was stretched along the Earth line and also flattened by its spin; the crown of the dome did not contract as a free tip (a requirement, not a derived result; Paper 7).

• **Data:** The crater densities on the six axes (Paper 7, §A.9); the anti-Earth point, $221$ after the mask.

• **Method:** A calculation of the tidal figure and its relaxation while the body rose.

• **Where in the series:** Paper 7, Further work item 6, §§E.2, A.9, D.2; Paper 1, §F.2.

## 61 The spacings between polar craters

The polar reading holds that the craters kept their size while the surface between them contracted; the direct measure of this is the distance from each crater to its nearest neighbour, by diameter class.

• **The question:** Did the gaps between the polar craters close, measured directly as nearest-neighbour distances, by diameter?

• **What the model predicts:** The craters kept their size while the surface between them contracted (Paper 7).

• **Data:** The Robbins catalogue.

- **Method:** Nearest-neighbour statistics, the caps against the equatorial belt.
- **Where in the series:** Paper 7, Further work item 7, §D.2.

## 62 The marks of the shell

A stiff shell contracting at the poles produces shortening structures (lobate scarps and wrinkle ridges), and a shell stretched at the equator produces extensional structures (graben); their distribution in latitude outside the maria is a test independent of the craters.

- **The question:** Are the shortening structures concentrated toward the poles and the extensional structures toward the equator, outside the maria?
- **What the model predicts:** As the stiff shell of Paper 7, §C.7 requires (a test independent of the craters; Paper 7).
- **Data:** Catalogues of lobate scarps, wrinkle ridges and graben.
- **Method:** Examining the distribution of the tectonic structures by latitude.
- **Where in the series:** Paper 7, Further work item 8, §C.7.

## 63 The innermost caps

The polar contraction must be greatest in the last $10^\circ$ of latitude, but the existing counts ($86$ craters in the north and $36$ in the south) are too small to decide this.

- **The question:** How does the crowding change within $10^\circ$ of each pole, where the contraction should be greatest?
- **What the model predicts:** The largest excess lies at the poles; the present counts are too small ($86$ craters in the north, $36$ in the south; Paper 7).
- **Data:** The next generation of catalogues reaching below $20\ \mathrm{km}$ diameter.
- **Method:** Building profiles within $10^\circ$ of each pole.
- **Where in the series:** Paper 7, Further work item 10, §D.4.

## 64 The shallower polar craters

Polar craters of $20 - 100\ \mathrm{km}$ diameter are shallower than equatorial craters (a depth-to-diameter ratio $d/D$ of about $0.062$ against $0.074$), and the shortening of the shell around them remains to be separated from later degradation.

• **The question:** Are the polar craters of $20 - 100$ km shallower ($d/D$ of about $0.062$ against $0.074$) because the shell shortened around them, or because of later degradation?

• **What the model predicts:** Not separated in the series (Paper 7).

• **Data:** LOLA depths; degradation classes.

• **Method:** A search for shortening structures around the craters against ordinary degradation.

• **Where in the series:** Paper 7, §E.2.

## 65 The alignment of elongated craters at middle latitudes

At latitudes $30^\circ - 60^\circ$, $45\% - 51\%$ of the most elongated craters ($e > 1.2$) are aligned north–south, against $20\% - 25\%$ elsewhere; it remains to be confirmed that the effect is real and does not come from a bias of low sun illumination.

• **The question:** Is the north–south alignment of the most elongated craters ($e > 1.2$) at $30^\circ - 60^\circ$ ($45\% - 51\%$ against $20\% - 25\%$) real, or an effect of low sun illumination?

• **What the model predicts:** Its size is a tenth or less of what deformation with the surface would give; it is measured, not derived (Paper 7).

• **Data:** Rim tracings from images lit from different directions.

• **Method:** An illumination check of the Robbins rims and a repetition with independently traced rims.

• **Where in the series:** Paper 7, §§E.2, D.5.

## 66 The polar scale: the onset of central peaks

The transition diameter between a simple bowl crater and a crater with a central peak depends on gravity and target strength, not on later contraction; the preserved diameters of the polar craters require this transition diameter to be the same at the poles and at the equator.

• **The question:** Is the crater diameter at which central peaks appear the same at the poles and at the equator?

• **What the model predicts:** The same transition diameter: the craters kept their size (Paper 7, §§D.2, D.5).

• **Data:** A catalogue of central-peak craters below $15$ km.

• **Method:** A count of central-peak craters by size in the caps and in the band.

- **Where in the series:** Paper 7, §D.5.

### 67 Two subjects left outside the model: crater ellipticity, and the poles against the marginal belt

Paper 7 examined two further questions and left each as a study in its own right, since each needs its own definitions of zone and of shape: one asks whether the contraction left any elongation (ellipticity) in the craters, and the other compares the polar caps not with the equator but with the marginal belt of Paper 13 (which, without its polar parts, holds about 149 craters per $10^6$ km², against 219 in the caps).

- **The question:** (a) Did the contraction leave any elongation in the craters? (b) How do the caps compare with the marginal belt of Paper 13 rather than with the equator (about 149 against 219 per $10^6$ km²)?
- **What the model predicts:** Both were examined in Paper 7 and left as studies in their own right, with their own definitions of zone and shape (Paper 7).
- **Data:** The Robbins catalogue.
- **Method:** Formulating independent definitions and counts.
- **Where in the series:** Paper 7, Further work item 4.

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PART G

# South Pole–Aitken

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In the LEF model the South Pole–Aitken (SPA) basin on the far side is not the scar of an external impact, but the far-side end of a heat path (“the burner”): while the Moon was attached to the tip of the pear, heat from the Earth’s interior entered through two exits on the near side (Imbrium as the main one and Serenitatis as the secondary), crossed the body and rose beneath the thin crust of the far side, where it raised a dome, compacted the column beneath it, closed its pores and erased part of the crater record (Paper 8). Topics 68–79 collect the open tasks for SPA (3 topics of physics and budgets that need new modelling, 7 tests that can be done on public data, and 2 topics that need new data).

**Topics of Part G**

| No. | Topic | Source | Readiness |
|---|---|---|---|
| 68 | A physical model of the heat axis and of its energy budget | Paper 8 §G | New modelling |
| 69 | A redox budget for the reduced mantle beneath SPA | Paper 8 §G, B12 | New modelling |
| 70 | The dynamic tests recomputed with the constants of the series | Paper 8 §D | New modelling |
| 71 | The cascade of volatiles and redox across the basin | Paper 8 §F | New data |
| 72 | Shallow craters and the depth minimum after the mare mask | Paper 8 §F | Public data |
| 73 | The null test for Serenitatis, and an out-of-sample test | Paper 8 §F | Public data |
| 74 | A large early mare in SPA, buried later | Paper 8 §F, B9 | New modelling |
| 75 | The late spray of small fragments | Paper 8 §F, B1h | Public data |
| 76 | Two crater populations and higher rims in young craters | Paper 8 §F, B1k | Public data |
| 77 | A direct porosity inversion for SPA | Paper 8 §G.1, B6 | Public data |
| 78 | The missing crust of SPA | Paper 8 §G.1, B4–B5 | Public data |
| 79 | Vapour that evaporated metals at SPA, and where it went | Paper 8 B11 | New data |

## 68 A physical model of the heat axis and of its energy budget

Paper 8 places the heat path along one north–south line (two exits on the near side, a dome and a hot centre on the far side), and what remains is to compute the energy budget this heat source delivered through the attached stage (about 600 Myr from the start of the bulge to release at 3.873 Ga, and at least ~$10^7$ years).

- **The question:** Was the heat that crossed the attached body from the exits of Imbrium and Serenitatis enough to compact the column, close the pores and erase the craters of SPA?
- **What the model predicts:** A single, long-lived heat source (“the burner”) working through the attached stage (Paper 8), supplying the energy needed for the work measured in Studies 1, 2 and 5: a crater density at the centre of 0.51 of that outside the margins (Study 1); a depth residual of 1.1 - 1.8 km at the centre, matching a column of 9 - 15 km that lost 12% of

its voids (Study 2); and porosity at 10 km depth closed to 2.9% - 4.4% at the centre (Study 5).

• **Data:** The results of Studies 1, 2 and 5 of Paper 8; the geometry of the path (exits, dome, hot centre) from Paper 8 Part C.

• **Method:** Building a model of heat transport along the path from the exits to the far-side crust, and computing the energy delivered through the attached stage against the energy needed for compaction and erasure.

• **Where in the series:** Paper 8 §G, Further work 1; Paper 1 §F.2.

## 69 A redox budget for the reduced mantle beneath SPA

Basalt fragments returned by the Chang'e-6 mission from the SPA basin show a markedly reduced oxygen fugacity of ΔIW -1.93, against -0.80 on the near side; the model attributes this to the prolonged action of the heat source driving off oxygen and to metallic iron from the Earth's interior taking up the remainder.

• **The question:** How much heat and how much metallic iron are needed to lower the oxygen fugacity from ΔIW -0.80 to -1.93, and is this consistent with the heat budget of topic 68?

• **What the model predicts:** Oxygen driven off by the heat and consumed by iron from the Earth's interior (Paper 8, B12), so that the heat and iron required fall within the supply of the heat path.

• **Data:** The Chang'e-6 basalt fragments (23 fragments measured by two methods).

• **Method:** A redox mass-balance calculation (the oxygen to be removed from the source region, the iron that will consume it, and the heat that will drive the process), cross-checked against topic 68.

• **Where in the series:** Paper 8 §G, Further work 2; Paper 8 B12.

## 70 The dynamic tests recomputed with the constants of the series

Part D of Paper 8 examined the dynamic conditions for the motion of material and heat from the Earth's interior to the far side (flow speed, transit time, energy, pressure, the potential barrier and its easing in a continuous body, the sign of the net force and the exit angle) before the constants of the series were locked; a repeat calculation with the constants of Papers 3 and 4 is needed.

• **The question:** With a day of 5.77 h, the corotation radius of that state and the parent mass of the series, what are the effective potential barrier and the base pressure for a body with the shape of Paper 3 (LEF Mechanism in Practice)?

• **What the model predicts:** The recalculation is required by Paper 8 §D.3a, and the model requires that the barrier be eased in a continuous body and that the base pressure suffice to drive the flow.

• **Data:** Nothing new (a recalculation with the constants of Papers 3 and 4).

• **Method:** Repeating the tests of speed, transit time, energy, barrier, sign and exit angle of Paper 8 Part D under the constants of the series.

• **Where in the series:** Paper 8 §G, Further work 3, and §D.1–D.5; Paper 1 §F.2.

## 71 The cascade of volatiles and redox across the basin

If the SPA basin was heated from its centre, the loss of volatile elements and the reduction must be strongest near the crater minimum and fade toward the margins (Chang'e-6 samples taken about 210 km from the crater minimum already show a mantle that lost its volatiles on heating, Paper 8, B11).

• **The question:** Are the loss of volatiles and the reduction strongest near the crater minimum (45.4°S, 197.9°E) and weaker toward the margins?

• **What the model predicts:** A heated centre, in which the signature fades outward along the same radial profile as the crater deficit (Paper 8 Part F, row 2).

• **Data:** Orbital maps of thorium (Th), iron oxide (FeO) and hydroxyl (OH), sampled at the minimum and in rings around it.

• **Method:** Building radial profiles on the same rings as the crater study of Paper 8 (Study 1), under the same masks.

• **Where in the series:** Paper 8 Part F, row 2.

## 72 Shallow craters and the spatial depth minimum after the mare mask

Study 1 found that craters of 40 - 80 km diameter in the central 20° of SPA are 18% shallower than those of the marginal belt; to show that the shallowing comes from the burner, which compacted and raised their floors, and not from later lava fill, the depths must be examined after removal of the maria and the hidden maria (cryptomaria).

• **The question:** Do shallow craters persist outside the buried basalts, and where are the craters shallowest once the maria and their haloes are removed?

• **What the model predicts:** The shallowing comes from the burner and not from lava fill (Paper 8); shallow craters should remain where there is no lava, and the depth minimum should coincide with the crater minimum (Part F, row 3).

• **Data:** Crater depths from the Lunar Orbiter Laser Altimeter (LOLA); maps of maria and cryptomaria.

• **Method:** Mapping crater depth under the mare mask (removing the maria, the cryptomaria and a halo around each) and locating the depth minimum.

• **Where in the series:** Paper 8 Part F, row 3.

## 73 The null test for Serenitatis, and an out-of-sample test

The crater minimum of SPA lies on the longitude of the antipode of Serenitatis (197.9° against 198.8°); it remains to test by a permutation test whether this match is statistically rare among all the near-side openings, and to confirm it on data not used in building the model.

• **The question:** Would a deficit placed as precisely toward an antipode arise by chance for any opening? Does the reading hold on data not used in building it?

• **What the model predicts:** The deficit follows the antipodes of the near-side openings, Imbrium and Serenitatis (Paper 8); among all the openings the match is rare by chance, and a deficit unrelated to the antipodes would leave the burner without its place (Part F, row 6).

• **Data:** The catalogue of openings (Paper 5, Appendix A); published crater catalogues.

• **Method:** Running a permutation over all the openings (computing the antipode of each opening and measuring the rank of the match of Serenitatis to the deficit), followed by an out-of-sample test on a crater catalogue or size range not used in Study 1.

• **Where in the series:** Paper 8 Part F, row 6.

## 74 A large early mare in SPA, buried later

The model reads the mare patches visible inside SPA as the tops of a larger early mare that closed and was covered by material that landed later, so that the buried basalt extends beyond the visible patches and the load of its weight shows in the Bouguer gravity.

• **The question:** Do the buried basalts extend well beyond the visible patches, and does the gravity field show the load of a buried basalt fill?

• **What the model predicts:** A large early mare that closed and was buried (Paper 8, B9, A.5); if the buried basalts end at the visible patches and the gravity field shows no buried fill, there was no large early mare (Part F, row 7).

• **Data:** Dark-halo craters (small craters that excavate dark basalt through a bright cover); spectra from the Moon Mineralogy Mapper ($M^3$); GRAIL Bouguer gravity.

• **Method:** Mapping buried basalts from dark-halo craters and spectra, and building a gravity model of the basalt fill, its thickness and its extent.

• **Where in the series:** Paper 8 Part F, row 7; B9; A.5.

## 75 The late spray of small fragments

In the model the late stages of the attached body threw out a spray of small fragments that landed as small, fresh craters on top of old, large craters; after the first test found a ratio of small to large craters of $4.62$ inside SPA against $2.13$ elsewhere (B1h), the second test is frozen and ready to run.

• **The question:** Are the small craters on large host craters inside SPA fresh while their hosts are degraded?

• **What the model predicts:** A late spray of small fragments (Paper 8); the frozen second test examines 20 host craters of $40 - 80$ km diameter inside SPA and 20 in a control area at the same latitudes, with $2,634$ candidate small craters ranked on a fixed freshness scale (depth relative to fresh craters of the same size; Krüger et al. 2018; Part F, row 8).

• **Data:** Lunar Reconnaissance Orbiter Narrow Angle Camera (LROC NAC) images.

• **Method:** Classifying and ranking each crater by the fixed criteria before comparing SPA with the control area.

• **Where in the series:** Paper 8 Part F, row 8; B1h.

## 76 Two crater populations and higher rims in young craters inside SPA

The history of SPA has two parts: an old crater record partly erased by the heat, and young craters formed in a compacted target (closed pores), which therefore keep higher rims (as

measured for craters of $40 - 80$ km at the centre, B1k); this topic separates the craters by age.

- **The question:** Are the old craters degraded and the young ones fresh, and do young craters inside SPA have higher rims than matched young craters outside it?
- **What the model predicts:** Two populations of different age, and young craters formed in a compacted target (Paper 8; Part F, rows 9–10).
- **Data:** Crater ages (degradation state or counts on the ejecta) and rim heights from LOLA.
- **Method:** A matched comparison inside SPA and outside it, by age and by size.
- **Where in the series:** Paper 8 Part F, rows 9–10; B1k.

## 77 A direct porosity inversion for SPA

Study 5 inferred, by extrapolation from published gravity inversions, that the porosity at $10$ km depth is almost entirely closed at the centre of SPA ($2.9\% - 4.4\%$); a direct regional inversion for SPA itself would remove the dependence on the chosen depth model.

- **The question:** Does an inversion made for SPA itself confirm almost entirely closed porosity at $10$ km depth, and most strongly at the centre?
- **What the model predicts:** At $10$ km depth, a porosity of $2.9\% - 4.4\%$ at the centre rising to $7.1\% - 8.8\%$ at the margins (against $6.2\% - 12.8\%$ around), while at the surface SPA is as porous as the far side around it (Paper 8, B6).
- **Data:** GRAIL gravity; LOLA topography.
- **Method:** A regional inversion of density and porosity with depth on rings centred on the basin.
- **Where in the series:** Paper 8 §G.1 (limitation of Study 5); B6.

## 78 The missing crust of SPA

Study 3 found an outer ring of thickened crust ($3.8 - 5.7$ km at $50^\circ - 55^\circ$ from the centre, in seven of eight directions) that holds about $37\%$ of the crust missing from SPA, and Study 4 found an inner ring in the west; the full volume balance remains to be closed.

- **The question:** Where is the rest of the crust missing from SPA, given that the outer ring holds about $37\%$ of it?

• **What the model predicts:** The crust slid off the rising dome and piled up at the margins, and part of it lies in the inner ring in the west (Paper 8, B4, B5); the volume of the thinned basin should balance against the two rings and the regional trend of the far side.

• **Data:** The four GRAIL crustal models.

• **Method:** A volume-balance calculation of the crust missing from the basin against the volume in the outer ring, in the inner ring and in the regional trend.

• **Where in the series:** Paper 8 §G.1 (limitation of Study 3); B4; B5.

## 79 Vapour that evaporated metals at SPA, and where the vapour went

Paper 8 (B11) found a low, flat $\mathrm{K/Th}$ ratio in the heated region ($451 - 455$ against $563$ in the far-side highlands and $537$ in the rest of the far side) and heavy isotopes of potassium ($\delta^{41}\mathrm{K}$ of $0$ to $+0.09$) and iron ($\delta^{56}\mathrm{Fe}$ of $0.13 - 0.21$) in the Chang'e-6 basalts; the author proposes that hot water vapour from a cooked magma carried volatile metals with it as chlorides or hydroxides, and the other metals and the fate of the vapour remain to be measured.

• **The question:** Is the mantle source beneath SPA very dry, and are zinc, copper, sodium and chlorine depleted and isotopically heavy together with potassium and iron? Where did the vapour go: to space, or did it condense at the margins?

• **What the model predicts:** The mantle source is very dry; $\mathrm{K}$, $\mathrm{Na}$, $\mathrm{Zn}$, $\mathrm{Cu}$, $\mathrm{Fe}$ are depleted together and isotopically heavy, and the chlorine is isotopically heavy; vapour that recondensed at the margins left an excess of volatiles and light isotopes there (Paper 8, B11, B12).

• **Data:** Chang'e-6 samples (water content and isotopes of $\mathrm{K}$, $\mathrm{Fe}$, $\mathrm{Zn}$, $\mathrm{Cu}$, $\mathrm{Cl}$) against Apollo and Chang'e-5 basalts; findings of condensed $\mathrm{Na}$ and $\mathrm{K}$ in Chang'e-6 glasses (Li et al. 2026); the $\mathrm{K/Th}$ ratio at the margins ($567$; Paper 8, B11).

• **Method:** Measuring the volatile metals and chlorine as one set (to distinguish this from mixing of materials), and comparing the isotopes of the condensed phases with the residue and the margins to close the vapour budget.

• **Where in the series:** Paper 8, B11, B12, B6 (related to topics 68, 69 and 71).

> RESULT
>
> **Part G in summary:** The heat path of Paper 8 was measured in five studies; what remains open is the physics of the path (68–70), the remaining tests of Part F (71–76),

> the two stated limitations (77–78) and the budget of vapour and metals (79); six of the twelve topics can be done at once on public data.

PART H

# The stripes and the Pacific

The floor of the North Pacific is crossed by long, straight, parallel stripes that run east–west for thousands of kilometres (from Mendocino in the north to Clipperton in the south). Paper 9 reads these stripes as the folded leading side of the pear on which the Moon was built: the pear stood with its base near Kiritimati (1.9°N, 157.5°W) and leaned back against the Earth's rotation, so that its front was stretched and folded into stripes as it collapsed, and its back crumpled into the arcs of the western Pacific. Topics 80–90 collect the 9 studies of Paper 9 and the two topics of the deep Earth and the Pacific lithosphere (9 of them can be done at once on public data, 1 needs new data and 1 needs new modelling).

### Topics of Part H

| No. | Topic | Source | Readiness |
|---|---|---|---|
| 80 | Study 1 — the equator refitted on full traces | Paper 9 §G | Public data |
| 81 | Study 2 — the chance test of the base on the equator | Paper 9 §G | Public data |
| 82 | Study 3 — the stripes on one side only | Paper 9 §G | Public data |
| 83 | Study 4 — steps in depth and age across each stripe | Paper 9 §G | Public data |
| 84 | Study 5 — the trailing side | Paper 9 §G | Public data |
| 85 | Study 6 — continuity beneath the ridges | Paper 9 §G | Public data |
| 86 | Study 7 — the spatula ridges against ridges built from below | Paper 9 §G | Public data |
| 87 | Study 8 — the mirror imprint | Paper 9 §G | Public data |
| 88 | Study 9 — the stripes beneath the dated layers | Paper 9 §G | Public data |
| 89 | The Pacific province as the returned pear (R10, R11) | Papers 3, 13 | New data |
| 90 | An independent reconstruction of the Pacific lithosphere | Paper 13 | New modelling |

## 80 Stripes Study 1 — the equator refitted on full traces

Parallel stripes lying along small circles about a common pole define the equator of the stripe system, which is inclined to today's equator (11° in the first fit, crossing today's equator at 159°W, and up to about 20° in a refit on the author's own points) and passes through the base of the pear.

- **The question:** What is the inclination of the equator of the stripes, and where does it cross today's equator, with errors?
- **What the model predicts:** One equator, inclined at least 10° (first fit 11°, and up to about 20°), passing through the base of the pear (Paper 9).
- **Data:** Bathymetry (GEBCO); the full traces of the stripes drawn on it.
- **Method:** Fitting small circles about a common pole to the traces along each stripe, with errors computed from the scatter of the points.
- **Where in the series:** Paper 9 §G, Study 1, and §E.1–E.3; Paper 1 §F.3, item 6.

## 81 Stripes Study 2 — the chance test of the base on the equator

The base of the pear near Kiritimati lies 1.6° from the equator of the stripe system; a Monte Carlo test will determine whether this closeness is statistically rare.

- **The question:** How often does a point placed at random in the region fall within ± 2° of the equator of the system?
- **What the model predicts:** The base lies on the equator of the system, 1.6° from it (Paper 9, B.2).
- **Data:** Bathymetry (GEBCO) and the full traces of the stripes (as in topic 80).
- **Method:** Running a Monte Carlo simulation to estimate how often random points fall within ± 2° of the fitted equator, over the range of fits of topic 80.
- **Where in the series:** Paper 9 §G, Study 2; §B.2.

## 82 Stripes Study 3 — the stripes on one side only

Since the stripes represent the leading side (the front) that was stretched and folded east of the base, the region west of the base must show bending, convergence and crumpling of the back of the pear, and not a continuation of the same stripes.

• **The question:** Do the stripes begin at the base and run east, and is what lies west of it bending and crumpling rather than a continuation of the same stripes?

• **What the model predicts:** The stripes are the leading side of the pear (Paper 9, B.3, B.6).

• **Data:** Bathymetry; maps of seafloor magnetic anomalies.

• **Method:** Following each stripe westward from the base and recording where it ends, bends or converges.

• **Where in the series:** Paper 9 §G, Study 3; §B.3, B.6.

## 83 Stripes Study 4 — steps in depth and age across each stripe

If the stripes are folds of one coherent structure (the leading side of the pear), the steps in the depth and age of the seafloor across each stripe must be ordered systematically about the base and not at random.

• **The question:** Does the seafloor change in steps of depth and age across each stripe, in an order about the base?

• **What the model predicts:** An order about the base, which ties each stripe to the pear (Paper 9).

• **Data:** Bathymetry; seafloor age grids.

• **Method:** Building sections across each stripe and reading the step in depth and age as a function of distance from the base.

• **Where in the series:** Paper 9 §G, Study 4.

## 84 Stripes Study 5 — the trailing side

The back of the pear lay west of the base, and as the pear collapsed it crumpled into the arcs of the western Pacific (at a distance of 8,100 - 10,300 km from the base).

• **The question:** Is the Pacific asymmetric east–west about the base, with parallel stripes on the leading side and crumpled arcs on the trailing side?

• **What the model predicts:** The back of the pear collapsed into the arcs of the western Pacific, 8,100 - 10,300 km from the base (Paper 9, B.8).

• **Data:** Bathymetry; the geometry of the arcs.

• **Method:** An east–west comparison about the base of the shape, spacing and direction of the structures at matching distances.

- **Where in the series:** Paper 9 §G, Study 5; §B.8; §H; Paper 1 §D.1.

## 85 Stripes Study 6 — continuity beneath the ridges

If the stripes are an early fold of the leading side, they must pass beneath the seamount chains and ridges that cross them without a break on the same circle (four crossings were observed in Paper 9), and where a fold formed beneath a standing ridge the ridge should be shortened, compressed or bent.

- **The question:** Does each stripe continue on the same circle on both sides of a ridge, and is the ridge shortened where the fold passed beneath it?
- **What the model predicts:** The stripes pass beneath the ridges (four crossings observed; Paper 9, B.5–B.6, F.2).
- **Data:** Bathymetry.
- **Method:** Measuring the coordinates of each stripe and the length of the ridge on a flat map without vertical exaggeration (to avoid the distortions of an oblique view).
- **Where in the series:** Paper 9 §G, Study 6; §B.5–B.6, F.2; Paper 1 §D.1.

## 86 Stripes Study 7 — the spatula ridges against ridges built from below

Paper 9 reads the three chains Line Islands, Marshall–Gilbert–Tuvalu and Hawaiian–Emperor as ridges drawn upward from above by the raised tip at spacings of 2,800 - 3,500 km, as distinct from plume ridges built from below (Ninetyeast, Walvis and Chagos–Laccadive).

- **The question:** Do the chains Line, Marshall–Gilbert–Tuvalu and Hawaiian–Emperor differ from Ninetyeast, Walvis and Chagos–Laccadive in age, shape, flexural moat, crustal root and direction? How does the age progression of Hawaii fit in?
- **What the model predicts:** The chains were drawn upward from above by the raised tip at spacings of 2,800 - 3,500 km; Line and Marshall show no plume age progression, while the age progression of Hawaii–Emperor (0 - 81 Ma) is a late sequence of eruptions on a ridge that already stood (Paper 9, D.1–D.4, F.3).
- **Data:** Radiometric ages; gravity and flexure data (flexural moat); seismic measurements of crustal roots.
- **Method:** A multi-dimensional comparison of the six chains on the five measures, and an examination of the Hawaiian ages as late eruptions on an existing ridge.

• **Where in the series:** Paper 9 §G, Study 7; §D.1–D.4, F.3, §H.

## 87 Stripes Study 8 — the mirror imprint

When the Moon sat on the tip of the pear, its near side faced the Earth and left on the floor beneath it a mirror image enlarged $1.37$ times (because of its low density then — $1,300\ \mathrm{kg/m^3}$ against $3,344\ \mathrm{kg/m^3}$ today): with Imbrium in its place ($13.19^\circ\mathrm{N}, 179.04^\circ\mathrm{E}$), the mirror predicts Serenitatis near $11.5^\circ\mathrm{N}, 166.3^\circ\mathrm{E}$ and the centre of the contact near $8.2^\circ\mathrm{N}, 179.3^\circ\mathrm{W}$.

• **The question:** Are circular features found on the floor at the mirrored positions of Serenitatis, Nectaris, the centre of the contact and the other near-side maria, at the scale $1.37$?

• **What the model predicts:** A mirror image of the near side, with Imbrium in its place (Paper 9, C.3).

• **Data:** Bathymetry; gravity.

• **Method:** Marking the expected positions of each near-side mare on the map before the check, and examining them for circular features against random control positions.

• **Where in the series:** Paper 9 §G, Study 8; §C.3.

## 88 Stripes Study 9 — the stripes beneath the dated layers

The seafloor of the Pacific is dated to at most about $180\ \mathrm{Myr}$, whereas the pear fell back at release $3.873\ \mathrm{Ga}$ ago; the model reconciles the two in that the dated seafloor is the last young layer laid over an ancient fold beneath it.

• **The question:** Does the fold continue beneath the youngest layers, with thinner layers over its crests and thicker layers in its troughs?

• **What the model predicts:** The dated seafloor (at most about $180\ \mathrm{Myr}$) is the last layer laid on an earlier fold (Paper 9, F.1).

• **Data:** Seismic reflection sections; drill holes of the Deep Sea Drilling Project and its successors (DSDP, ODP, IODP).

• **Method:** Measuring the thickness of each layer across the fold on seismic sections and drill holes that cross a stripe.

• **Where in the series:** Paper 9 §G, Study 9; §F.1; §H; Paper 1 §D.1.

## 89 The Pacific province as the returned pear (R10, R11)

When the Moon separated, the rest of the pear's mass ($7 - 12$ lunar masses, about nine lunar masses — about $11\%$ of the Earth's mass — wholly within the corotation radius with no remnant in orbit, requirement R10) collapsed to the bottom of the mantle beneath the base of the pear and formed the low-velocity province (LLVP) of the Pacific (requirement R11), while the African province is the undisturbed opposite lobe; with the position test passed in 5 of 5 tomographic models at $2,800$ km depth beneath Kiritimati, the composition, the shape and the isotopes of the plumes remain to be tested.

- **The question:** Is the Pacific province richer in iron, denser, lower and broader than the African one? Is its northern part the base of the pear? What is its relation to the ultra-low-velocity zone (ULVZ) of Hawaii? Do the plumes rising from it carry only the Earth's signature?
- **What the model predicts:** An iron-rich returned pile beneath the base of the pear; $7 - 12$ lunar masses returned whole with no remnant in orbit (R10); the base above the northern part of the province, about $1,500$ km north of its centre, and plumes from it carrying a terrestrial signature only (Papers 3 and 13).
- **Data:** Tomographic models; density constraints from the Earth's normal modes; isotopes of plume basalts ($\mathrm{He}$, $\mathrm{W}$, $\mathrm{Ne}$, $\mathrm{Xe}$).
- **Method:** Comparing the two provinces in density and shape; an isotopic survey of the plumes of the Pacific and of the African province.
- **Where in the series:** Paper 3 §E.1–E.4; Paper 13 §C.5, C.6 (R10, R11); Paper 1 §F.3 item 5, §D.1.

## 90 An independent structural reconstruction of the Pacific lithosphere

The series places the source region in the Pacific from the reading of the Moon (the base near Kiritimati, the back of the pear near $25°\mathrm{N}, 165°\mathrm{E}$, and the tip at $9.3°\mathrm{N}, 169.3°\mathrm{E}$, about $3,760$ km west of the base); a structural and palaeomagnetic reconstruction of the lithosphere made blind, without the lunar map, would confirm the match independently.

- **The question:** Does a reconstruction of the lithosphere beneath the Pacific, made without the lunar map, place the source region where the series reads it?

• **What the model predicts:** The base near Kiritimati; the back of the pear near $25^{\circ}\mathrm{N}, 165^{\circ}\mathrm{E}$; the tip at $9.3^{\circ}\mathrm{N}, 169.3^{\circ}\mathrm{E}$ (Papers 3, 9 and 13).

• **Data:** Published reconstructions of the Pacific lithosphere; palaeomagnetic data.

• **Method:** Making an independent reconstruction blind to the lunar map and comparing it with the map afterwards.

• **Where in the series:** Paper 13, Further work 7, and §C.2.

> RESULT
>
> **Part H in summary:** The nine studies of Paper 9 (topics 80–88) can be done at once on public data; topic 89 (the Pacific province as the returned pear) has passed the position test and awaits a check of composition and plume isotopes; topic 90 closes the part with an independent reconstruction of the lithosphere.

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PART I

# Mars and other bodies

Paper 10 applies the principles of the model to the Mars system: a young Mars, 1.5 times its present radius and completing one rotation in 5 hours, fired two crustal plugs from a single volcanic conduit at Pavonis Mons, pushed by trapped gas; a ring of condensed steam supplied the sideways speed they lacked, rounded their orbits and drew them into the equatorial plane. The Japanese MMX mission (landing on Phobos in 2027 and returning samples in 2031) will decide some of the predictions. Topics 91–106 collect the 16 research questions of this part (9 need new modelling, 4 need new data and 3 can be done at once on public data).

### Topics of Part I

| No. | Topic | Readiness |
|---|---|---|
| 91 | The expanded, five-hour young Mars and the loss of its spin | New modelling |
| 92 | The gas push in a long conduit: the critical step | New modelling |
| 93 | The soft plug that seals: the cycle that builds it | New modelling |

| | | |
|---|---|---|
| 94 | Drag in the conduit and the braking of Phobos | New modelling |
| 95 | The ring of condensed steam: mass, structure and coupling | New modelling |
| 96 | The tidal climb of Deimos and a soft young Mars | New modelling |
| 97 | The race of Phobos with the corotation line | New modelling |
| 98 | The samples of MMX: Martian material, one source, vesicles | New data |
| 99 | Pits and grooves on Phobos | Public data |
| 100 | The depth of the hollows on Deimos and Phobos | Public data |
| 101 | The centre of mass of Phobos | New data |
| 102 | The four calderas of Tharsis and the mass deficit beneath Pavonis | Public data |
| 103 | An ancient component in the Martian dust | New data |
| 104 | One conduit on the Martian equator, and the water beneath Tharsis (R18) | New data |
| 105 | The other bodies: Venus, Mercury and fast-spinning super-Earths | New modelling |
| 106 | The height difference between the poles and the spin at formation | New modelling |

## 91 The expanded, five-hour young Mars and the loss of its spin

Paper 10 assumes that the young Mars was 1.5 times its present radius (5,084 km) with a day of 5 hours (a free equatorial sideways speed of 1.78 km/s, an effective gravity of 1.04 m/s² and a corotation line at 7,057 km), and so it had to lose about 92% of its spin by shedding volatiles through the corotation line.

- **The question:** Can a self-consistent hot, volatile-rich young Mars have this radius and spin, and can it lose about 92% of its spin by shedding 5% - 20% of its mass through the line?
- **What the model predicts:** A radius of about 1.5 times today's and a day of 5 hours; a loss of 5% - 20% of the initial mass (mostly volatiles; the lower value if the density rose toward the centre; Paper 10).
- **Data:** Models of the Martian interior constrained by InSight; models of early volatile loss and of the deuterium enrichment of Martian water.
- **Method:** A coupled model of the thermal contraction, the volatile loss and the spin of the young Mars, with the angular momentum carried by mass leaving through the corotation line.

- **Where in the series:** Paper 10 §B.1–B.4, §H.3, Part O item 1.

## 92 The gas push in a long conduit: the critical step

To reach speeds of 2.4 - 2.9 km/s (beyond the ~1 km/s limit of a free volcanic explosion), Paper 10 relies on trapped gas pushing a sealing plug along a confined conduit of 200 - 300 km at a mean pressure of 0.25 - 0.47 GPa (balanced by a surrounding confining pressure of 0.26 - 0.39 GPa) and with an energy of $1.3 \times 10^{22}$ J for Deimos and $3.6 \times 10^{22}$ J for Phobos.

- **The question:** Can gas trapped in a deep chamber push a plug of crust along a conduit of 200 - 300 km to 2.4 - 2.9 km/s without breaking the walls or bypassing the plug?
- **What the model predicts:** A mean driving pressure of 0.25 - 0.5 GPa balanced by the confining pressure along the path, and a gas mass of 1 to 3 times the plug mass ($10^{16}$ - $3 \times 10^{16}$ kg of steam; Paper 10).
- **Data:** The rheology and strength of the Martian crust and mantle at depth; the volatile contents of Martian magmas.
- **Method:** A one-dimensional and then two-dimensional model of a gas-driven plug in a deformable conduit, computing together gas expansion, wall stress and leakage past the plug.
- **Where in the series:** Paper 10 §C.3, §C.6, §C.8; Part O item 3.

## 93 The soft plug that seals: the cycle that builds it

The plug in Paper 10 is a hot, highly viscous body that seals the conduit a few kilometres below the exit vent and is built in a Vulcanian cycle (Clarke et al. 2015) until its complete break-out (as the inward-folded impact marks on both moons attest).

- **The question:** In what range of viscosity and temperature does a plug of Martian crust seal a conduit against 0.25 - 0.5 GPa and then leave as one body instead of breaking into fragments?
- **What the model predicts:** A soft, sealing plug that leaves whole; the first plug (Deimos) small, light and wet, and the second (Phobos) larger, drier and harder (Paper 10).
- **Data:** Viscosity and fracture data for crystal-rich, volatile-bearing magmas; terrestrial volcanic sequences on the Earth.
- **Method:** A model of the cooling, sealing and fracture of a viscous plug in a conduit over repeated cycles.

- **Where in the series:** Paper 10 §C.4–C.5, §D.4.

## 94 Drag in the conduit and the braking of Phobos

A plug moving up a conduit on a rotating planet is pressed sideways by the Coriolis effect ($2\omega v \approx 1.8$ - $2.0$ m/s²) against the western wall, which accounts for the grooves of Phobos and its rapid braking.

- **The question:** How much speed and how much mass does a plug lose against the conduit wall, and is the grooving concentrated on the side that faced west?
- **What the model predicts:** Phobos was braked to 2.4 - 2.6 km/s and lost about 1% of its mass, and its grooves are concentrated on the side that faced west (while Deimos left without drag; Paper 10).
- **Data:** The groove maps of Phobos; the shape model.
- **Method:** A contact-and-friction model of a viscous plug in a rotating conduit, and an examination of the direction of the grooves against the predicted western side.
- **Where in the series:** Paper 10 §C.7, §G.3, §K.1.

## 95 The ring of condensed steam: mass, structure and coupling

A plug fired from the surface lacks the sideways speed needed for a Keplerian orbit and gains it by friction in a ring of condensed steam (a hot gap inside, ice crystals and dust in the middle, and clumps beyond the Roche limit for ice at ~13,000 km).

- **The question:** Can a young Mars build and keep such a ring, and does a body crossing it gain its orbit, round it and draw it into the equatorial plane before it is ground away?
- **What the model predicts:** A ring of ~$10^{16}$ kg; Deimos was ground by 30% - 60% of its mass in the outer ring, while Phobos was ground by about 1% at the edge of a fading ring (Paper 10).
- **Data:** Models of circumplanetary discs and of the atmosphere and water of early Mars.
- **Method:** An N-body or hydrodynamic simulation of a plug crossing a ring of condensed steam, including drag, grinding and the damping of the orbit to the equator.
- **Where in the series:** Paper 10 §E.2, §F.1–F.3, §H.1; Part O item 4.

## 96 The tidal climb of Deimos and a soft young Mars

The tide pushed Deimos from the ring (~15,000 km) to its present orbit (23,463 km), beyond the present corotation line (20,428 km), before the line reached there — a process that requires dissipation 8 - 20 times higher in the young Mars.

- **The question:** What history of $k_2/Q$ for a hot, soft, fast Mars carries Deimos to 23,463 km before the corotation line reaches 20,428 km, given its larger exit mass?
- **What the model predicts:** Tidal dissipation 8 - 20 times today's over 0.5 - 1 Gyr (Paper 10).
- **Data:** The present orbit of Deimos; models of dissipation in a partly molten planet.
- **Method:** A coupled integration of the spin rate of Mars, its dissipation and the orbit of Deimos.
- **Where in the series:** Paper 10 §G.2, §J.1; Part O item 2.

## 97 The race of Phobos with the corotation line

Phobos was fired to a height of 10,000 - 11,000 km and pushed outward at first, but the corotation line, receding faster as Mars slowed, caught it near 19,000 km (at a day of ~22 hours), and Phobos has been falling inward since.

- **The question:** Does one history of the spin of Mars and of its dissipation give both the outward push of Phobos and the faster outward motion of the line, so that the line catches Phobos near 19,000 km?
- **What the model predicts:** The line catches Phobos near 19,000 km at a day of about 22 hours; running its fall backward at today's dissipation ($k_2/Q = 1.816 \times 10^{-3}$) places it at 19,100 - 19,500 km 3.9 - 4.5 Gyr ago, in agreement with Cheng et al. (2022); the inclination fell from 1.48° to today's 1.09° (Paper 10).
- **Data:** The present orbit and inclination of Phobos; the measured dissipation of Mars.
- **Method:** The same coupled integration as in topic 96, run for Phobos, including the damping of the inclination.
- **Where in the series:** Paper 10 §G.3, §K.4–K.5; Part O item 5.

## 98 The samples of MMX: Martian material, one source, vesicles

The MMX mission will return more than 10 g of regolith from the surface of Phobos in 2031; in the LEF model Phobos is a Martian crustal plug, hydrothermally altered by Martian water, from

a single planetary source, never melted in a giant impact nor recondensed from a disc.

- **The question:** Is Phobos Martian crust from one source, or a mixture of Mars and an impactor, or primitive asteroidal material?
- **What the model predicts:** Oxygen, chromium and titanium isotopes falling on the Martian fractionation line; no component of an external impactor; zinc and potassium isotopes not shifted by evaporation; hydrated minerals present; and closed gas vesicles of a magmatic foam that cooled (Paper 10, N.1).
- **Data:** The regolith samples to be returned by the MMX mission (2031).
- **Method:** Multi-system isotopic analysis and textural microscopy of the returned grains.
- **Where in the series:** Paper 10 §N, Appendix B.

## 99 Pits and grooves on Phobos

In the model the grooves and the pit chains of Phobos are one structure formed in the conduit, and the grooves were cut by gas-borne dust, not smoothed by water.

- **The question:** Do pits ever cross grooves? Are the groove walls V-shaped and the edges sharp or only slightly degraded?
- **What the model predicts:** Pits never cross grooves; V-shaped groove walls; sharp or slightly degraded edges, unlike the rounded surface of Deimos (Paper 10, N.2).
- **Data:** Existing images and shape models of Phobos; MMX imaging from 2027.
- **Method:** Mapping the pit chains against the grooves; cutting profiles across the groove walls and edges.
- **Where in the series:** Paper 10 §K.1–K.3, §N.

## 100 The depth of the hollows on Deimos and Phobos

The shape models measured for Paper 10 (Ernst et al. 2023) give a median depth-to-width ratio of the hollows of 1.9% on Deimos against 5.5% on Phobos (Deimos is much more worn); both moons have a sharp end and a blunt end facing Mars.

- **The question:** Does a complete catalogue of the hollows on both moons confirm the contrast, and does it show that the rims of Deimos are worn rather than buried?
- **What the model predicts:** Shallow, worn hollows on Deimos, and deeper ones on Phobos; low ridges on Deimos that are the roots of its highest folds (Paper 10, N.3).

- **Data:** Public shape models of both moons; MMX passes near Deimos.
- **Method:** Automatic and manual measurement of the depth, width and rim profile of every hollow above a size threshold.
- **Where in the series:** Paper 10 §J.1–J.2, §L.

## 101 The centre of mass of Phobos

A plug that cooled from its surface and was turned by tidal forces so that its blunt end faces Mars need not be uniform in its internal density.

- **The question:** Is the centre of mass of Phobos offset from its centre of figure toward Mars, toward its blunt end?
- **What the model predicts:** An offset toward Mars and toward the blunt end (while the usual explanations give no such prediction; Paper 10, N.5).
- **Data:** Close MMX tracking of Phobos.
- **Method:** An inversion of the gravity field compared with the shape model.
- **Where in the series:** Paper 10 §N; Appendix B.

## 102 The four calderas of Tharsis and the mass deficit beneath Pavonis

The summit pit of Pavonis is a single circle 47 km wide and almost 5 km deep (depth-to-width ratio ~0.10), with a volume of 8,300 - 8,700 km³ — of the order of the two moons together (6,740 km³) — read in the model as the roof of a chamber that collapsed after the firings.

- **The question:** Is Pavonis the deepest caldera relative to its width among the four volcanoes of Tharsis, and is there a mass deficit beneath it?
- **What the model predicts:** The deepest relative to its width (~0.10), with a mass deficit beneath it (Paper 10, N.4).
- **Data:** MOLA topography; gravity models of Mars.
- **Method:** Computing the depth and width of each caldera from MOLA and a local gravity inversion beneath each.
- **Where in the series:** Paper 10 §I.1–I.3, §N.

## 103 An ancient component in the Martian dust

In the model the inner part of the ring fell back on Mars as dust near the equator, so its grains, having stayed in space, carry solar-wind gases and a composition identical to the regolith of Phobos.

- **The question:** Does Martian dust contain an ancient component of 4 Gyr, with grains carrying solar-wind helium and neon and the composition of the regolith of Phobos?
- **What the model predicts:** Such an ancient component, most common in the broad equatorial band and in the dust continents (Tharsis, Arabia Terra, Elysium), mixed with material ground by drought and wind (Paper 10, N.7).
- **Data:** Returned Perseverance regolith samples; MMX samples of Phobos for comparison.
- **Method:** Grain-by-grain dating and noble-gas analysis, compared with the Phobos samples.
- **Where in the series:** Paper 10 §H.1, §N; Appendix B.

## 104 One conduit on the Martian equator, and the water beneath Tharsis (R18)

Pavonis Mons (1.48°N) is the nearest to the equator of the four volcanoes of Tharsis (against Olympus Mons at 18.65°N), in keeping with the orbital inclination of the two moons (< 2° from the equator); the water that did not leave remained sealed beneath the crust (as with the water-saturated middle crust found by InSight).

- **The question:** Are the two moons Martian material from one equatorial conduit, and is a water-saturated middle crust found beneath Tharsis as well?
- **What the model predicts:** One equatorial source for both moons (with Deimos the wetter of the two), and a water-saturated middle crust beneath Tharsis as beneath InSight (Paper 10, N.5–N.6; Paper 13, R18).
- **Data:** MMX samples and passes near Deimos; future seismic stations on Tharsis.
- **Method:** Comparing the composition of the two moons and seismic sounding of the crust of Tharsis.
- **Where in the series:** Paper 10 §D.1–D.4, §H.2, §M; Paper 13 §C.7 (R18).

## 105 The other bodies: Venus, Mercury and fast-spinning super-Earths

Venus (rotation in 243 d) and Mercury (rotation in 59 d) spin too slowly and dry out too fast near the Sun (their corotation radius lies beyond the Hill radius), and so lack moons, whereas a massive, fast-spinning super-Earth is the opposite case.

- **The question:** What does a parameter study in spin, mass and volatile content predict for the moons of other planets?
- **What the model predicts:** Nothing is raised on slowly spinning bodies (Venus, Mercury); more on a fast-spinning super-Earth (Paper 1).
- **Data:** The satellite inventory of the planets; the spins and masses of exoplanets.
- **Method:** Applying the corotation and pressure tests of the series across the parameter space.
- **Where in the series:** Paper 1 §F.3 item 9; Paper 10 §B; Paper 11 §§C, D, L.4.

## 106 The height difference between the two poles and the spin at formation

On Mars the north pole is several kilometres lower than the south pole, as part of the Martian dichotomy; the author's proposal ties this difference to the spin rate of the body at its formation, since the spin rate sets how close the corotation radius lies and how much a young body can raise (Papers 2, 10).

- **The question:** Does the height difference between the north pole and the south pole of a body grow with the rate at which the body spun at its formation?
- **What the model predicts:** The faster the spin at formation, the larger the difference between the poles (a hypothesis to be tested, not an existing result of the series).
- **Data:** Polar topography from orbital altimetry (MOLA for Mars — Smith et al. 1999; LOLA for the Moon; and corresponding data for further bodies); the spin of the young Mars from Paper 10 (§§B.1–B.2) and the spins of the series for the Earth and the Moon.
- **Method:** In a first stage, modelling the one-sided component that turns fast spin into a polar difference (since spin alone flattens both poles equally); in a second stage, measuring the difference on the ground beneath the polar ice caps relative to the equipotential surface against the spin at formation (taking into account possible polar wander due to the load of Tharsis).
- **Where in the series:** Raised in the author's working notes (collection of 16.10.2026); related to topics 104, 105 and 129.

PART J

# Moon births in the solar system

Paper 11 applies the mechanism of the model to the whole solar system: moons are born from the parent at the corotation line, in a time window set by the parent's stage of differentiation, in one of three modes — built (our Moon, the only one), ejected (Mars to Neptune) or split (Pluto and Charon). Topics 107–120 collect the questions left open in Parts M and N of Paper 11 (4 can be done at once on public data, 6 need new modelling, and 4 need data from future missions).

### Topics of Part J

| No. | Topic | Readiness |
|---|---|---|
| 107 | The arc of births: moon production by distance from the Sun | Public data |
| 108 | Mercury and Venus: no bulge, no birth | Public data |
| 109 | How much water escaped through the line: the line at the time of birth in the giants | New modelling |
| 110 | The inner body of the giants against their moons | New modelling |
| 111 | The density order of the mid-sized moons of Saturn | Public data |
| 112 | The density order of the moons of Uranus | New data |
| 113 | The 98° tilt of Uranus | New modelling |
| 114 | Grooves as the mark of a forced exit from a narrow opening | Public data |
| 115 | The breaking of Pluto and Charon and its scars | New data |
| 116 | The small moons of Pluto: before the split or at the moment of breaking | New modelling |
| 117 | Split pairs: the larger partner is denser | New data |
| 118 | The capture rule at the edge: Nereid and Himalia | New modelling |
| 119 | The dwarf planets: Haumea, Ceres, dwarf or moon | New data |
| 120 | The equatorial ridge of Iapetus | New modelling |

## 107 The arc of births: moon production by distance from the Sun

The extent of moon production depends on the planet's distance from the Sun: near the Sun the planet dries out fast and the window closes; farther out the planet is rich in water and ejects more moons; and beyond Saturn the surfaces freeze and produce fewer moons (the ratio of the mass of the regular moons to the parent rises from Jupiter, $2.1 \times 10^{-4}$, to Saturn, $2.5 \times 10^{-4}$, and falls at Uranus, $1.0 \times 10^{-4}$, and at Neptune, about $5 \times 10^{-7}$).

- **The question:** Do the mass of the regular moons relative to the parent, and their number, form an arc by distance from the Sun with a peak at Saturn, when all the regular moons are counted with measured masses?
- **What the model predicts:** An arc: no birth at Mercury and Venus; one built moon at the Earth; two small moons at Mars; a rise to a peak at Saturn (about 24 regular moons, $2.5 \times 10^{-4}$); a fall at Uranus (19, $1.0 \times 10^{-4}$) and at Neptune (7, about $5 \times 10^{-7}$) (Paper 11).
- **Data:** The masses of the regular moons (Jacobson 2010, 2014); the masses of the inner moons of Neptune have no direct measurement and are estimated from size and an assumed density.
- **Method:** Building a uniform catalogue of regular moons after removing the captured ones by the capture rule (Paper 11 §L.3), and measuring the mass of Proteus and the inner moons of Neptune in a future mission.
- **Where in the series:** Paper 11 §L.4, Figure L.3; §A.10 item 4.

## 108 Mercury and Venus: no bulge, no birth

In the model Mercury and Venus dried out fast near the Sun, and the soft layer beneath the surface was sealed before a birth could take place (and today their corotation radius lies beyond the Hill radius: for Venus $1.5 \text{ million km}$ against $\approx 1 \text{ million km}$, and for Mercury $\approx 243,000 \text{ km}$ against $\approx 220,000 \text{ km}$), unlike the proposal of Alemi & Stevenson (2006) that Venus bore a moon and lost it.

- **The question:** Are the surfaces of Mercury and Venus free of signs of a past bulge or pear?
- **What the model predicts:** No signs of a bulge; Aphrodite Terra is not a remnant of a bulge (Paper 11).
- **Data:** Topography and gravity from Magellan (Venus) and MESSENGER (Mercury); in future VERITAS, EnVision and BepiColombo.

• **Method:** A joint analysis of topography and gravity in search of a long equatorial plateau with an excess or deficit of mass matching a bulge.

• **Where in the series:** Paper 11 Parts C and D, §L.4, Part N.

## 109 How much water escaped through the line: the line at the time of birth in the giants

At Uranus and Neptune the corotation line today lies far outside the birth range ($3.24$ and $3.39$ radii); the model holds that the line migrated outward as water escaped through it, carrying angular momentum with it ($2\% - 3\%$ of the mass of Uranus and Neptune as a lower bound).

• **The question:** How much water had to escape through the line at each giant to bring the line from its place at the time of birth to its place today?

• **What the model predicts:** At the time of birth the line stood within the birth range of the ejection mode (between $1.74$ and $2.06$ radii); at Uranus and Neptune an escape of at least $2\% - 3\%$ of the mass is required (Paper 11).

• **Data:** The masses, radii and moment-of-inertia factors of the giants; the orbits of the moons.

• **Method:** A reverse angular-momentum balance with contraction inside and swelling outside (Paper 11 §A.5) along the history of differentiation, using the radius and the moment-of-inertia factor at the time of birth.

• **Where in the series:** Paper 11 §A.3, §A.5, Parts I and J; Part M item 5.

## 110 The inner body of the giants against their moons

The moons of the giants were born from the inner body of the parent and not from the hydrogen–helium envelope, so the rule “the parent is denser than its moons” compares the densities of the moons with the density of the inner body without the gas envelope. At the same time the envelope acts as a lid that sets how much water remained in the upper layer from which the moons were ejected.

• **The question:** What is the density of the inner body of each giant, with the hydrogen–helium envelope taken out of the account, and is it higher than that of each of its regular moons?

• **What the model predicts:** The inner body is denser than every regular moon: at Saturn at least 1,880 kg/m³ (the density of Titan), and at Uranus at least 1,711 kg/m³ (the density of

Titania) (Paper 11).

• **Data:** The gravity fields of the Juno and Cassini missions; interior models of the giants; the densities of the moons.

• **Method:** Building interior models that separate the inner body from the gas envelope and comparing its density with the densities of the moons.

• **Where in the series:** Paper 11 §A.9 (density without the gas envelope, Figure A.5); Parts G–J.

## 111 The density order of the mid-sized moons of Saturn

In the Saturn system the mid-sized moons are not ordered by density as a function of distance from the planet: Mimas (1,149 kg/m³), Enceladus (1,609 kg/m³), Tethys (984 kg/m³), Dione (1,476 kg/m³) and Rhea (1,237 kg/m³). In the model these moons were born in stage 2 of the differentiation as clumps of water that gathered into moons above the Roche limit.

• **The question:** What sets the density order of the mid-sized moons of Saturn within stage 2?

• **What the model predicts:** Open; the model proposes checking whether Saturn's thick lid caused intermittent feeding of the opening (Paper 11 Part M item 6).

• **Data:** The densities, frozen shapes and surfaces of the mid-sized moons from the Cassini mission.

• **Method:** Comparing the density with the frozen shape and the surface age of each moon, as evidence of the place and time of its birth.

• **Where in the series:** Paper 11 Part H; Part M item 6; Part N (Saturn).

## 112 The density order of the moons of Uranus

At Uranus the density rises with distance (unlike Jupiter: Miranda $1,200$, Ariel $1,592$, Umbriel $1,390$, Titania $1,711$, Oberon $1,630\ \mathrm{kg/m^3}$), which is explained by the stages of differentiation (Titania and Oberon left at its start and reflect the parent, and Miranda left at its peak, laden with water and of low density).

• **The question:** Does the degree of differentiation of each moon of Uranus match the stage at which it was born?

• **What the model predicts:** Titania and Oberon are less differentiated and reflect the parent; Miranda is rich in water (Paper 11).

• **Data:** Moment-of-inertia factors and gravity fields of the moons (requiring an orbiter at Uranus).

• **Method:** Gravity measurement during a close pass of each moon by a space mission to Uranus.

• **Where in the series:** Paper 11 Part I; Part M item 1; Part N (Uranus).

## 113 The 98° tilt of Uranus

The origin of the axial tilt of Uranus ($98^\circ$) remains open: water escape alone requires $117\%$ of the original angular momentum, and shifting the centre of mass alone does not tilt the axis, so a calculation of the effect of the distribution of mass is needed (moons and water on a distant orbit, or an early moon that returned onto the heavy side); the resonances in the literature have also been considered (Boué & Laskar 2010; Rogoszinski & Hamilton 2020; Saillenfest et al. 2022).

• **The question:** Can a change in the distribution of the mass of Uranus — moons and water on a distant orbit, or an early moon that left and returned — lead to a tilt of $98^\circ$?

• **What the model predicts:** Open; it was decided not to decide (Paper 11 §I.1).

• **Data:** The masses of the moons, the coefficient $J_2$ of Uranus, the conditions of the early system.

• **Method:** An integration of the spin axis with an effective $J_2$ of moons and water in orbit, and of a moon returning onto the heavy side.

• **Where in the series:** Paper 11 §I.1, Figures I.1–I.3; Part M item 2.

## 114 Grooves as the mark of a forced exit from a narrow opening

A moon ejected by force from a narrow opening under internal pressure leaves parallel grooves (a straight exit, as in the grooves of Phobos) or twisted grooves and rings at the end (a screwing exit, as in the belts of Miranda). In the literature the grooves of Phobos are explained as material thrown from craters on Mars (Murray & Heggie 2014), and the belts of Miranda as material that rose from within (Pappalardo et al. 1997).

• **The question:** Does the geometry of the grooves on Phobos and of the belts on Miranda match the mark of a forced exit from a narrow opening, straight or screwing?

• **What the model predicts:** Parallel grooves along a direction of exit, or twisted grooves and rings at the end (Paper 11).

• **Data:** The groove maps of Phobos (HRSC, HiRISE); Voyager 2 images of Miranda.

• **Method:** Mapping the geometry of the grooves and belts and comparing them with the two patterns of exit.

• **Where in the series:** Paper 11 Part I (grooves); Part F.

## 115 The breaking of Pluto and Charon and its scars

Pluto split by the breaking of a frozen crust over a soft interior: on Charon the rough region in the north (Oz Terra) is the original crust and the smooth plain on the equator facing Pluto (Vulcan Planitia, Beyer et al. 2017) is the interior that was exposed and rounded; on Pluto the scar migrated before locking according to the sign of its mass.

• **The question:** Where is the scar of the break on Pluto — near the pole, or is Sputnik Planitia the filled hollow of the break — and what is the origin of the second smooth plain on the far side of Charon?

• **What the model predicts:** The surfaces of Pluto exposed in the break also smoothed in the rounding; the place of the scar is set by the sign of its mass (Paper 11).

• **Data:** New Horizons images (including the smooth plain about $400\ \mathrm{km}$ across on the far side of Charon, Beyer et al. 2021, and imaging of the side of Pluto facing Charon).

• **Method:** Geological mapping of the side facing Charon and a calculation of the wander of Pluto's spin axis with a scar of mass deficit or excess.

• **Where in the series:** Paper 11 §K.1; Part M item 3; Part N (Pluto).

## 116 The small moons of Pluto: before the split or at the moment of breaking

The four small moons of Pluto (Styx, Nix, Kerberos and Hydra) were ejected from the equator and the neck on circular orbits, and it remains to decide whether all of them left before the split or some were ejected at the moment of breaking.

• **The question:** Did all the small moons of Pluto leave before the split, or did some of them leave at the moment of breaking?

• **What the model predicts:** Ejection before the split, on circular orbits in the plane of Charon (Paper 11).

• **Data:** The orbits, sizes and surfaces of the small moons (Showalter & Hamilton 2015; Weaver et al. 2016).

• **Method:** A simulation of ejection from the equator of a body on the verge of separation against ejection at the moment of breaking, and a comparison with the orbits and the ratios of orbital periods.

• **Where in the series:** Paper 11 §K.1; Part M item 11.

## 117 Split pairs: the larger partner is denser

In a pair formed by splitting, the larger lobe keeps the core and so is denser than the smaller one (such as Pluto and Charon: $1,854$ against $1,702\ \mathrm{kg/m^3}$; Eris and Dysnomia: $2,400$ against $\approx 700\ \mathrm{kg/m^3}$; and Triton as the larger lobe of a split pair captured by Neptune).

• **The question:** In all the large binary pairs of the Kuiper belt, is the larger partner denser than the smaller?

• **What the model predicts:** Yes: the larger partner is denser (Paper 11).

• **Data:** Masses and sizes of binary pairs (Orcus and Vanth are not yet decisive, Brown & Butler 2023).

• **Method:** Orbit and size observations to determine the density of each partner in a pair.

• **Where in the series:** Paper 11 Parts J and K.

## 118 The capture rule at the edge: Nereid and Himalia

By the scoring rule fixed in advance in Paper 11 (retrograde orbit = 2 points; large inclination, high eccentricity, unusual distance and different composition = one point each; capture threshold $\geq 3$ points), Nereid and Himalia score only 2 points and so remain to be checked as born of the parent.

• **The question:** Are the eccentric orbit of Nereid, and the distance of Himalia and its group, consistent with birth from the parent and an outward push?

• **What the model predicts:** Both were born from the parent, by the rule (Paper 11 §L.3).

• **Data:** The orbits and composition of Nereid and of the Himalia group.

- **Method:** An orbital integration from birth at the corotation line with the pushes the model describes.
- **Where in the series:** Paper 11 §L.3; Part N (Jupiter, Neptune).

## 119 The dwarf planets: Haumea, Ceres, dwarf or moon

Haumea (rotation in $3.9$ hours, the ends of the long axis beyond the corotation line by $1,104$ km) is read as an elongated body that has not yet split and may be shedding material now, while Ceres (corotation line at $2.54$ radii) lacks a moon.

- **The question:** Is Haumea losing material from its ends now, what blocked a birth at Ceres, and what distinguishes a dwarf from a moon that escaped?
- **What the model predicts:** Haumea: material leaves from the ends, and its family, ring and moons are tied to it; Ceres: the window, the size or the nearness to Jupiter blocked the birth (Paper 11).
- **Data:** The shape, ring and moons of Haumea; Dawn data for Ceres.
- **Method:** Observations of material near the ends of Haumea; a model of the birth window at Ceres.
- **Where in the series:** Paper 11 §K.2, §K.5, §K.6; Part M items 7–9.

## 120 The equatorial ridge of Iapetus

On the equator of Iapetus (and also on Pan and Atlas) there is a prominent equatorial ridge; Iapetus was born in stage 0 of the differentiation as a clump of water and froze at 2.53 Saturn radii (above the Roche limit), and the question is whether the ridge formed from material drawn out or plastered on at separation from the parent, like the spatula/trowel ridges of Paper 9.

- **The question:** Did the equatorial ridge of Iapetus, and of Pan and Atlas, form at the time of separation from the parent?
- **What the model predicts:** Open (Paper 11 Part M item 10).
- **Data:** Cassini topography and images of Iapetus, Pan and Atlas.
- **Method:** Building a model of material drawn out or plastered on along the equator at separation, and comparing it with the height and width of the observed ridge.
- **Where in the series:** Paper 11 Part H; Part M item 10; Paper 9.

PART K

# Requirements and key tests

Paper 13 sets the model against eighteen requirements (R1–R18), each matched with the measurement that tests it. Topics 121–132 collect the tests and balances left open there, headed by the two deciding key observations: the geometry of exposure (topics 121–122) and the source of the craters in returned terrestrial material (topic 123).

### Topics of Part K

| No. | Topic | Readiness |
|---|---|---|
| 121 | T1 crustal thickness rerun on the current geometry, all four GRAIL models | Public data |
| 122 | The belt statistics completed: T2 by age, T3, T4, GRGM1200B | Public data |
| 123 | No foreign material: the isotopic decision (R14) | New data |
| 124 | A multi-system isotope balance (R1) | New modelling |
| 125 | The metal budget: the core as a gauge (R2, R13) | New modelling |
| 126 | The joint crust–centre-of-mass solution (R3, R7) | New modelling |
| 127 | Two crusts, two ages (R5) | New data |
| 128 | Volatiles in the amount and phase the feed needs (R6) | New modelling |
| 129 | North–south asymmetry with South Pole–Aitken removed | Public data |
| 130 | The composition of the lunar core | New data |
| 131 | The core offset as the residual of the centre-of-mass offset | New modelling |
| 132 | An orphan minimum of crustal thickness | Public data |

## 121 T1 crustal thickness rerun on the current geometry, all four GRAIL models

The model requires the crust to be thinnest over the contact (the area exposed last), thicker in the belt around it, and thickest on the far side (exposed from the start); after the geometry was updated to a contact centre at 24.1°N, 335.2°E (cap 0° - 31°, belt 31° - 70°, far part 70° - 180°), Model 1 was run and gave a contrast of 9.9 km (7.5 km without the maria and SPA),

and Models 2–4 were left for a rerun; the author ruled on 15.10.2026 that this rerun is left to other researchers.

- **The question:** With the full GRAIL grids for all four models, on the current geometry, does the ordering cap < belt < far hold, and how large is the far-minus-contact contrast, as a number for calibration?
- **What the model predicts:** The crust thickens monotonically from the cap to the belt to the far side (rank correlation +0.87); the contrast on the September geometry was 9.65 / 10.0 / 10.3 / 10.6 km, and on the current geometry Model 1 gives 9.9 km (7.5 km after removal of the maria and SPA), while Models 2–4 are left for a rerun (Paper 13).
- **Data:** The four GRAIL crustal-thickness models (Wieczorek et al. 2013) at full resolution.
- **Method:** Computing belt means by angular distance from the centre of the contact, with the zones fixed in advance and the same mask applied in every zone.
- **Where in the series:** Paper 13 §B.8, §B.9, Further work item 1, §D.3 (R8); Paper 1 §D.1, §E.2.

## 122 The belt statistics completed: T2 by age, T3, T4, GRGM1200B

After test T5 (the thickness trend in 12 belts, correlation +0.87) and the general test T2 (a far/margin crater-density ratio of 1.34) passed, what remains is to run T2 on old craters sorted by age (degradation state), to run the basin tests T3 and T4, and to examine the ordering on independent gravity grids (Bouguer, lateral density and GRGM1200B).

- **The question:** Is the oldest crater population ordered by belt when it is selected by age? Do at least 60% of the ringed mascon basins lie at 41° - 105° (T3)? Do at least 70% of the ten strongest mascons without rings lie within 50° (T4)? Does the ordering hold on the Bouguer and lateral-density grids?
- **What the model predicts:** Three concentric zones about the centre on the near side, and no other axis (Papers 3 and 13); T5 (+0.87) and T2 (1.34) passed, and T3 and T4 have not yet been run.
- **Data:** The Robbins (2019) crater catalogue with degradation states; GRAIL gravity gradients; the GRGM1200B model.
- **Method:** Carrying out the checks as in Paper 13 §B.8, with thresholds fixed in advance before the data are examined.

• **Where in the series:** Paper 13 §B.8 (T2–T5), Further work item 1; Paper 3 §D.1; Paper 1 §F.3 item 1, §E.2, §D.1.

## 123 No foreign material: the isotopic decision (R14)

In the model about 99% of the lunar craters were made by returned terrestrial material, so the metal trapped in the old samples when they formed must carry a terrestrial nucleosynthetic isotopic fingerprint (unlike external asteroidal bodies, which would also have left in the upper 5 km of the crust 4.4 - 5.0 × $10^{19}$ kg of foreign mass — 2% - 6%, against at most 0.9% - 2% actually measured).

• **The question:** Does the metal that was part of the old samples when they formed — the iron–nickel of the two dated norites of Chang'e-6, and the siderophile elements in the melts of every basin of the maria — carry a terrestrial or a non-terrestrial isotopic signature (Cr, Ti, Mo, Ru, Os)?

• **What the model predicts:** About 99% of the craters are made of returned terrestrial material (Papers 1, 5, 6, 8, 13); the norite metal is terrestrial (Paper 8, B14, partly fitting; the isotopes decide).

• **Data:** The Chang'e-6 norites (CE6C0100YJFM002-158 and CE6C0200YJFM001EGP07-16); melts from crater sites of Apollo and Chang'e; Pacific plume basalts as a terrestrial reference.

• **Method:** Nucleosynthetic isotope analyses and a comparison with a mixing model of a terrestrial returning body with a terrestrial target against asteroidal patterns.

• **Where in the series:** Paper 8 B14, §E.1–E.2, Part F row 1, §G.2 item 8; Paper 5 §J.7, Further work item 4, §M.3; Paper 6 §L.5, Further work item 6, §O.1; Paper 13 §C.6 (R14), §D.3; Paper 1 §E.2, §F.3 item 4, §D.1.

## 124 A multi-system isotope balance (R1)

The Earth and the Moon are almost identical in oxygen isotopes ($\Delta^{17}O$ = -1 ± 5 ppm), but the model requires that one initial composition (the mantle of the early Earth plus the deep upwelling that fed the pear) close at the same time the isotopic balance of titanium, chromium, tungsten and volatile elements after separation and evaporation.

• **The question:** Does one composition — the mantle of the early Earth plus the deep upwelling, after separation and evaporation — match the lunar data in O, Ti, Cr, W and the

volatile elements together?

• **What the model predicts:** Terrestrial material throughout; oxygen: $\Delta^{17}O$ = -1 ± 5 ppm between the Earth and the Moon (Paper 13).

• **Data:** Published lunar and terrestrial isotope data.

• **Method:** Running a multi-system mixing and fractionation model with one set of inputs for all the systems.

• **Where in the series:** Paper 13, Further work item 2; §A.1, §B.1, §D.3; Paper 1 §E.1 (R1).

## 125 The metal budget: the core as a gauge (R2, R13)

The Moon's core is small: about 1.4% of its mass (range 1% - 2%), with a radius of about 330 km. In the model the Moon's material rose from a source next to the iron that descended to the Earth's core, but not from that iron itself; a rising foamed melt dragged droplets of that iron with it — some sank on the way and some were carried up — so the size of the core gauges how deep the feed reached.

• **The question:** What fraction of the descending iron does a rising foamed melt drag along, and does it give a lunar core of about 1.4% (1% - 2%) from a source next to the descending iron, but not from that iron itself?

• **What the model predicts:** The small core gauges how deep the feed reached (Papers 3 and 13).

• **Data:** The seismic size of the core (about 1.4%); the feed rate of Paper 3.

• **Method:** The fluid dynamics of droplet entrainment and settling in a rising melt.

• **Where in the series:** Paper 13, Further work item 3; §A.2, §B.2, §C.6 (R13), §D.3; Paper 3 §C.10; Paper 1 §F.2.

## 126 The joint crust–centre-of-mass solution (R3, R7)

The Moon's centre of mass is offset by about $2\ \mathrm{km}$ from its centre of figure toward the Earth; Paper 13 shows that a crust of the modelled shape offsets the centre of mass by $1.24 - 3.11\ \mathrm{km}$ for density contrasts of $300 - 750\ \mathrm{kg/m^3}$, and that the mass of the dome shell ($2.85 \times 10^{21}\ \mathrm{kg}$) matches the mass of the far-side crust ($2.6 \times 10^{21}\ \mathrm{kg}$) to about $10\%$. What remains is to replace this sensitivity range with a joint inversion of one supported density model that gives at the same time the size of the offset, its direction and the crustal thickness.

- **The question:** Does one supported density model give the size and the direction of the $2\ \mathrm{km}$ offset of the centre of mass, together with the measured crustal thickness, on the geometry of the exposure zones?
- **What the model predicts:** A sensitivity range of $1.24 - 3.11\ \mathrm{km}$ for density contrasts of $300 - 750\ \mathrm{kg/m^3}$ against the observed $2\ \mathrm{km}$ or so, and a shell mass of $2.85 \times 10^{21}\ \mathrm{kg}$ against the far-side crust of $2.6 \times 10^{21}\ \mathrm{kg}$ (Paper 13).
- **Data:** GRAIL gravity; LOLA topography; the GRAIL crustal models.
- **Method:** A joint inversion of crustal thickness and centre of mass.
- **Where in the series:** Paper 13, Further work item 4; §B.3, §B.7, Appendix A, §D.3; Paper 3 §D.2; Paper 1 §F.2.

## 127 Two crusts, two ages (R5)

Since in the model the surface of the far side was the outer part of the dome from the start and was preserved while the near side was attached to the Earth, the highland crust of the far side must be older than comparable highland crust on the near side (comparing crystallisation age with crystallisation age, outside impact basins).

- **The question:** Is the highland crust of the far side older than comparable highland crust on the near side, when the sampling is done outside the basins?
- **What the model predicts:** The far side closed and was preserved first (Paper 13).
- **Data:** Chang'e-6 material; future samples from the far side.
- **Method:** Comparing primary crystallisation ages (clock 1) of comparable highland units on both sides outside the basins.
- **Where in the series:** Paper 13, Further work item 5; §A.5, §B.5, §D.3.

## 128 Volatiles in the amount and phase the feed needs (R6)

Volatiles play two roles: dissolved or as bubbles they lower the density of the feed so that it can stand up to the corotation radius, and in expanding they supply the driving pressure; what remains is to link the water measured in lunar melt inclusions ($615 - 1,410\ \mathrm{ppm}$) with the amount and phase needed for density, rheology and pressure at the same time.

- **The question:** Does the water measured in lunar melt inclusions ($615 - 1,410\ \mathrm{ppm}$) match the amount and phase that the density, rheology and pressure of the feed require at

one and the same time?

• **What the model predicts:** Volatiles lower the density of the feed and supply its pressure (Papers 3 and 13).

• **Data:** Water measurements in melt inclusions and in apatite.

• **Method:** Cross-checking the transport requirement of topic 4 against the measured inventory.

• **Where in the series:** Paper 13 §A.6, §B.6, §D.3; Paper 3 §A.6.

## 129 North–south asymmetry with South Pole–Aitken removed

Since the South Pole–Aitken (SPA) basin lies wholly in the southern hemisphere and distorts the hemispheric means, testing whether a global north–south asymmetry exists requires full removal of the near–far asymmetry and of SPA and the large basins.

• **The question:** Does any north–south asymmetry of the Moon remain once the near–far asymmetry and the large basins, SPA first among them, are removed?

• **What the model predicts:** This must be tested, not assumed in advance; SPA alone may not confirm a hemispheric asymmetry (Paper 13).

• **Data:** The GRAIL crustal models; crater catalogues.

• **Method:** An analysis of north–south asymmetry with SPA and the large basins masked identically in both hemispheres.

• **Where in the series:** Paper 13 §C.5; Paper 7 §D.4.

## 130 The composition of the lunar core: the Earth's iron, caught on its way down

Planetary cores contain light elements dissolved in the iron that record the conditions of their formation (silicon and oxygen at the high pressures of a deep core, against sulphur at low pressure as droplets settle through a silicate mantle); since in the model the lunar core is terrestrial iron caught in the ascent while the Earth's iron was descending to its core (Paper 3, §C.10), its dominant light element should be sulphur and not silicon and oxygen.

• **The question:** Is the light element of the lunar core sulphur, and not silicon and oxygen?

• **What the model predicts:** The lunar core is terrestrial iron carried up with the feed while the Earth's iron was still descending (Paper 3, §C.10), and its light element is mainly sulphur;

a core whose light elements are silicon and oxygen would not support this reading.

• **Data:** The seismic core of Weber et al. (2011 — radius ~330 km, solid inner core ~240 km, density and wave speeds); seismic data from a future lunar geophysical network; laboratory measurements of density and sound speed in Fe–S and Fe–Si–O liquids at lunar-core pressures.

• **Method:** Comparing the measured density and wave speeds of the outer core with the candidate alloys under lunar conditions, and building a model of the light-element content of the terrestrial metal during its descent.

• **Where in the series:** Raised in the author's working notes (collection of 16.10.2026); related to Paper 3 §C.10, Paper 13 §A.2, §B.2, §C.6 (R2, R13) and topic 125.

## 131 The core offset as the residual of the centre-of-mass offset

While Paper 13 explains the offset of the lunar centre of mass (~2 km toward the Earth) by an asymmetric crust over a symmetric core (an offset of 1.24 - 3.11 km for a density contrast of 300 - 750 kg/m³), the author further proposes that the core itself (1% - 2% of the Moon's mass) is offset by 100 - 200 km or more toward the Earth, following the push of the burner and the pull of the Earth before it froze.

• **The question:** How much of the centre-of-mass offset, in size and direction, remains once the crust and the topography are taken into account, and does this residual fit a core offset toward the Earth?

• **What the model predicts:** A residual toward the Earth matching a core offset of the order of 100 - 200 km or more (a zero residual, or a residual pointing away from the Earth, leaves no room for an offset core).

• **Data:** LOLA topography (centre of figure); GRAIL gravity and the GRAIL crustal-thickness models; the seismic core (~330 km, about 1.4% of the mass).

• **Method:** Computing the full offset vector from the crust and the topography under one density model (topic 126), converting the residual vector into a displacement of a core of the seismic size, and computing the forces (tide, attachment and the push of the burner) that could have displaced the core and fixed it in place.

• **Where in the series:** Paper 13 §A.7, §B.7, Further work item 4; Paper 3 §E.5; the author's working notes (collection of 16.10.2026).

## 132 An orphan minimum of crustal thickness

In the model the thickness of the lunar crust is thinned only by the broad order of the exposure zones, beneath the openings (a jump of ≥ 8 km at 29 of 36 openings, Paper 5) and beneath the large basins; the existence of an “orphan” local minimum that lies beneath no opening and no basin would be a difficulty for the model.

- **The question:** Does the crustal-thickness map contain a local minimum that lies neither beneath a basin nor beneath an opening?
- **What the model predicts:** No such minimum exists; every local minimum of the crust lies beneath an opening or beneath a basin (and the broad thinning of the contact region is the order of the exposure zones).
- **Data:** The four GRAIL crustal-thickness models (Wieczorek et al. 2013); the openings of Paper 5; the catalogue of lunar basins identified in GRAIL gravity (Neumann et al. 2015).
- **Method:** An automatic search for local minima in all four models under depth and size criteria fixed in advance (such as the rings and the 8 km threshold of Paper 5 §D.2: centre to 0.3R against a ring at 1.3 - 2.0R) and cross-checking them against the openings and the basins.
- **Where in the series:** Paper 5 §D.2; Paper 13 §B.8; Paper 3 §B.7; the author’s working notes (collection of 16.10.2026).

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PART L

# How to choose

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This part sorts the 132 research topics into three working lists according to observational weight and to the tools and data available to the researcher.

## L.1 List 1 — the two key observations of the model

**1. The geometry of exposure (topics 121 and 122):** a concentric ordering of the crust and the craters about the centre of the near side (the contact cap, the marginal belt and the far part):

| Observational measure | Cap | Belt | Far part | Status and calibration values |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Craters ≥ 20 km per $10^6$ km² (current geometry, maria and SPA removed) | 53 | 187 | 220 | The ordering holds; far/belt: 1.17, far/cap: 4.15, belt/cap: 3.55 |
| Crustal thickness, Model 1 (km) (current geometry, maria kept) | 26.3 | 30.3 | 36.2 | The ordering holds; contrast: 9.9 km |
| Crustal thickness, Model 1 (km) (current geometry, maria and SPA removed) | 31.3 | 33.0 | 38.9 | The ordering holds; contrast: 7.5 km |

| Test T1: far-minus-contact contrast (km) | Model 1 | Model 2 | Model 3 | Model 4 |
|---|---|---|---|---|
| Contrast, September geometry | 9.65 | 10.0 | 10.3 | 10.6 |
| Ordering cap < belt < far | Holds | Holds | Holds | Holds |

RESULT

**The observation in summary:** The ordering of the crust and the craters holds in every model and in every count; the rerun of Models 2–4 on the current geometry (topic 121) and the completion of T2 by age, T3, T4 and the gravity grids (topic 122) will decide the observation.

**2. The source of the craters (topics 49 and 123):** the making of about 99% of the lunar craters by returned terrestrial material, as against external asteroidal bodies:

| Component of foreign material in the upper 5 km of the crust | Mass fraction |
|---|---|
| Held in the crust (measured, at most) | 0.9% - 2% |
| Required if the visible craters had been made by bodies from outside (4.4 - 5.0 × $10^{19}$ kg) | 2% - 6% |

RESULT

**The observation in summary:** The mass argument and the ruthenium measured so far support a terrestrial source; the analysis of the nucleosynthetic isotopes of the metal in the Chang'e-6 norites and in the melts of the mare basins (topic 123) and the reading of the melts against their source maria (topic 49) will decide the observation.

## L.2 List 2 — topics that can be done now on public data

These topics need no new mission or heavy simulation, but rest on existing public data (GRAIL gravity and crust, LOLA topography, LROC images, the Robbins 2019 crater catalogue, thorium and iron maps, bathymetry and age grids).

| Part | Topics that can be done at once on public data |
|---|---|
| B. The body on the tip | 19 The layered far-side crust; 21 KREEP as the torn edge of the sial; 23 Whole-Moon porosity by depth |
| D. The maria and the openings | 31 The order of sealing; 32 Thorium at high resolution over the feed domes; 33 The rim ring of the volatile-rich maria; 34 Gradual exposure within one mare; 35 Vents and conduits; 36 The titanium part of Study 13; 37 The collar of the open far maria; 38 The pits and caves of the Moon as vapour voids; 39 The crust jump by the age of the basin; 40 The diameter of an opening by the duration of its feeding; 41 The rim of a mare as a layered record; 42 Blunt or sharp crests on the side of the plain; 43 Gravity traces of the craters that vanished into the maria |
| E. The craters and their source | 51 The longitude record; 53 The Earth as the last scatterer; 54 Material poorer in iron after release; 55 Iron of the high and low floors; 56 Crater-count ages against direct ages; 57 The rays of young craters: material within the surface |
| F. The poles | 58 The young craters of the poles; 61 The spacing between the polar craters; 62 The marks of the shell; 64 The shallower polar craters; 65 The alignment of elongated craters; 66 The polar scale; 67 Crater ellipticity, and the poles against the marginal belt |
| G. South Pole–Aitken | 72 Shallow craters and the depth minimum; 73 The null test for Serenitatis; 75 The late spray of small fragments; 76 Two crater populations; 77 A direct porosity inversion; 78 The missing crust of SPA |
| H. The stripes and the Pacific | 80–88 Stripes Studies 1–9 |
| I. Mars and other bodies | 99 Pits and grooves on Phobos; 100 The depth of the hollows; 102 The four calderas and the mass deficit beneath Pavonis |
| J. Moon births across the solar system | 107 The arc of births; 108 Venus and Mercury: no bulge, no birth; 111 The density order of the mid-sized moons of Saturn; 114 Grooves as the mark of a forced exit |
| K. Requirements and key tests | 121 T1 rerun on the current geometry; 122 The belt statistics completed; 129 North–south asymmetry with SPA removed; 132 An orphan minimum of crustal thickness |

- **A note on priority:** Topics 121 and 122 appear in both List 1 and List 2, so the work that matters most for testing the model is also the most accessible to do at once.

## L.3 List 3 — topics that need a large simulation, a mission or samples

This list separates topics that need theoretical modelling or a large numerical simulation (in which the inputs are locked in advance by the series) from topics that need a space mission, returned samples or new laboratory experiments.

### 1. New modelling or a large simulation

| Part | Topics for modelling and simulation |
|---|---|
| A. The Earth before and during the pear | 1 Thermal expansion of the hot parent; 2 The stratified Maclaurin problem; 3 Core formation and the Hf–W history; 5 The energy of the lift; 6 The pressure test with the pear's mass; 7 The construction rate; 8 The pear's equilibrium figure (R9); 9 The northward lean of the pear; 10 The Earth's core during the pear; 11 The classical fission route, recomputed |
| B. The body on the tip | 12 3-D simulation of the connected body; 13 Stiff shell over a mobile base; 14 From a/c ≈ 1.20 to round; 15 The drift of the release latitude; 17 The trapped pressure at detachment; 18 Survival inside the fluid Roche limit; 20 The near-side crust over the contact |
| C. The orbit | 24 The coupled thermal–orbital integration; 25 The Earth's dissipation history; 26 The onset of recession; 27 The magma-ocean switch; 28 The fossil figure; 29 The early latitude, reconstructed |
| D. The maria and the openings | 30 The engine of the openings |
| E. The craters and their source | 44 The exit velocity; 45 Passage through the steam atmosphere; 46 Shock in a hot, layered target; 48 The crater size distribution, propagated; 50 Captured ejecta after release; 52 The trailing-side excess |
| F. The poles | 59 The figure with strength; 60 The tide on a body held at its base |
| G. South Pole–Aitken | 68 A physical model of the heat axis; 69 A redox budget for the reduced mantle; 70 The dynamic tests recomputed; 74 A large early mare in SPA, buried later |
| H. The stripes and the Pacific | 90 An independent reconstruction of the Pacific lithosphere |
| I. Mars and other bodies | 91 The young Mars and its spin; 92 The gas push; 93 The soft plug; 94 Drag in the conduit; 95 The ring; 96 The climb of Deimos; 97 The race of Phobos; 105 The other bodies; 106 The poles and the spin |

| | |
|---|---|
| J. Moon births across the solar system | 109 How much water escaped through the line; 110 The inner bodies of the giants; 113 The 98° tilt of Uranus; 116 The small moons of Pluto; 118 The capture rule at the edge; 120 The equatorial ridge of Iapetus |
| K. Requirements and key tests | 124 A multi-system isotope balance (R1); 125 The metal budget (R2, R13); 126 The joint crust–centre-of-mass solution (R3, R7); 128 The volatiles the feed needs (R6); 131 The core offset as the residual of the centre-of-mass offset |

### 2. A mission, samples or new laboratory data

| Part | Topics for space missions, samples and laboratory |
|---|---|
| A. The Earth before and during the pear | 4 The equation of state of the deep melt at 50 - 150 GPa |
| B. The body on the tip | 16 Stefan adhesion and failure of the contact; 22 High-pressure phases in samples from the contact zone |
| E. The craters and their source | 47 The trend in a series of events; 49 Seeing the returning body, not the ground |
| F. The poles | 63 The innermost caps |
| G. South Pole–Aitken | 71 The cascade of volatiles and redox across the basin; 79 Vapour that evaporated metals at SPA, and where it went |
| H. The stripes and the Pacific | 89 The Pacific province as the returned pear (plume isotopes) |
| I. Mars and other bodies | 98 The samples of MMX; 101 The centre of mass of Phobos; 103 The dust of Mars; 104 One conduit on the equator (R18) |
| J. Moon births across the solar system | 112 The density order of the moons of Uranus; 115 The breaking of Pluto and Charon and its scars; 117 Split pairs: the larger partner is denser; 119 The dwarf planets: Haumea, Ceres, and dwarf or moon |
| K. Requirements and key tests | 123 The isotopic decision (R14); 127 Two crusts, two ages (R5); 130 The composition of the lunar core |

- **The main weights in List 3:** Among the large simulations, topic 12 (the connected body in three dimensions) and topic 24 (the coupled thermal–orbital integration) carry the main weight; among the missions, the MMX mission decides four topics at once (topics 98, 100, 101 and 104, and also gives essential information for topic 99).

## FURTHER WORK Further work outside this paper

This section sets out the two complex fields of research named in the LEF series as future work, which have been left outside the list of 132 numbered topics in Paper 14.

• **The Earth–Moon axes system (from release to the present):** following the evolution of the four dynamical planes of the system — the Earth's equator, the Moon's orbit, the Moon's equator and the plane of the ecliptic — which move in two kinds of motion at different rates, under the influence of the shape memory of a frozen bulge. The first step in this research is to examine the channel of axial tilt: how the coupling of the Sun, the Earth and the Moon acted on the initial tilt of $3.8^\circ$ given to the Earth's spin axis at release (whether it enlarged or damped it on the way to the present value of $23.44^\circ$), and whether a change in the Earth's axis after separation caused the South Pole–Aitken basin to appear offset relative to the straight line facing the terrestrial burner.

• **The magnetism of the Moon:** the origin of the Moon's ancient magnetic field and its spatial pattern over the South Pole–Aitken basin (erased at the centre, strong on the trailing side, and including the magnetic swirls of Mare Ingenii), read in the series as a joint record of prolonged heating and of iron that came from the Earth. This study includes the test of the depth and direction of the magnetization, which must be strongest at the rim of the heated core region and come from a deep source.

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- LROC: Lunar Reconnaissance Orbiter Camera (WAC and NAC), NASA Planetary Data System.
- MMX: Martian Moons eXploration, Japan Aerospace Exploration Agency (JAXA).

**Use of AI tools:** The author used AI assistants (Claude, Anthropic; ChatGPT, OpenAI; Google's AI search assistant) to write and run the computation scripts, to analyse the catalogues and maps, to prepare figures and tables, to discuss the scenarios, and for language editing and translation. The model and its conclusions are the author's. The author checked the content and takes full responsibility for it.

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